

Robótica de Manipulação

Kinematics and Trajectory Planning

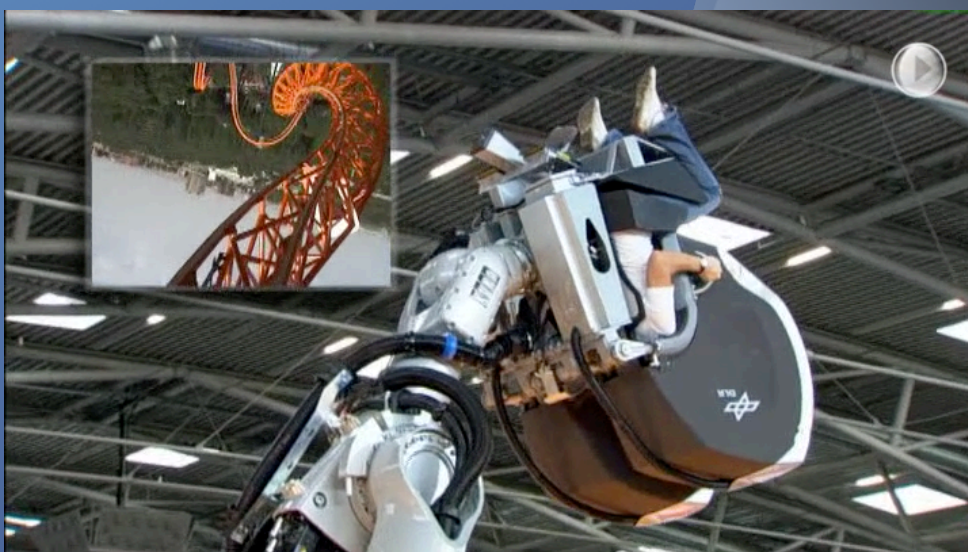
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KUKA Robocoaster
www.kuka-entertainment.com

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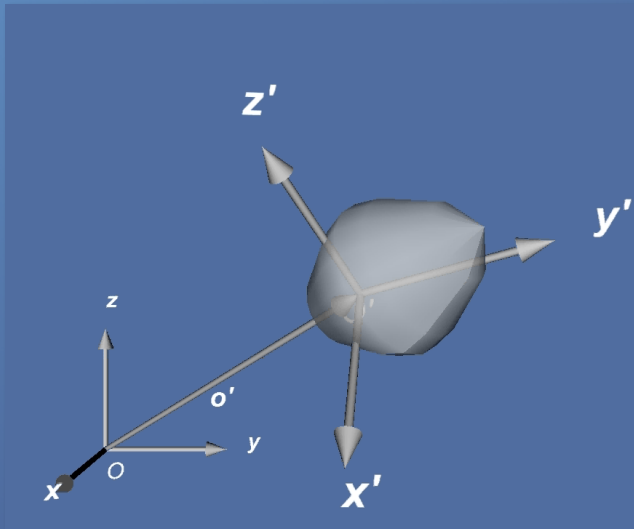


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Pose

A rigid body is completely defined in space by its **position** and **orientation** (**pose**) with respect to a reference frame



Base reference frame: $O - xyz$
 Body reference frame: $O' - x'y'z'$
 (both are orthonormal)

Position: $o' \in \mathbb{R}^3$
 $o' = o'_x x + o'_y y + o'_z z \rightarrow o' = \begin{bmatrix} o'_x \\ o'_y \\ o'_z \end{bmatrix}$

Orientation: *direction cosines*

$$\left. \begin{aligned} x' &= x'_x x + x'_y y + x'_z z \\ y' &= y'_x x + y'_y y + y'_z z \\ z' &= z'_x x + z'_y y + z'_z z \end{aligned} \right\} \rightarrow R = \begin{bmatrix} x'_x & y'_x & z'_x \\ x'_y & y'_y & z'_y \\ x'_z & y'_z & z'_z \end{bmatrix}$$



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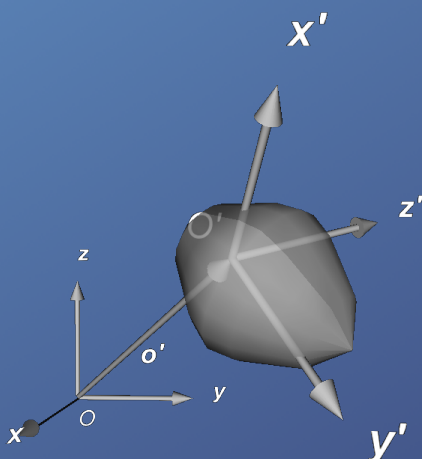
The Rotation Matrix

The rotation matrix belongs to the *special orthonormal group* $SO(3)$ of the real (3×3) matrices with orthonormal columns and determinant equal to 1

$$R = \begin{bmatrix} x'_x & y'_x & z'_x \\ x'_y & y'_y & z'_y \\ x'_z & y'_z & z'_z \end{bmatrix} = \begin{bmatrix} x'^T x & y'^T x & z'^T x \\ x'^T y & y'^T y & z'^T y \\ x'^T z & y'^T z & z'^T z \end{bmatrix} = [x' \quad y' \quad z']$$

Notable properties:

$$\begin{cases} R^T R = I_3 \\ R^T = R^{-1} \\ \det(R) = 1 \end{cases}$$



$$R^T = \begin{bmatrix} x'^T x & x'^T y & x'^T z \\ y'^T x & y'^T y & y'^T z \\ z'^T x & z'^T y & z'^T z \end{bmatrix} = \begin{bmatrix} x_{x'} & y_{x'} & z_{x'} \\ x_{y'} & y_{y'} & z_{y'} \\ x_{z'} & y_{z'} & z_{z'} \end{bmatrix}$$



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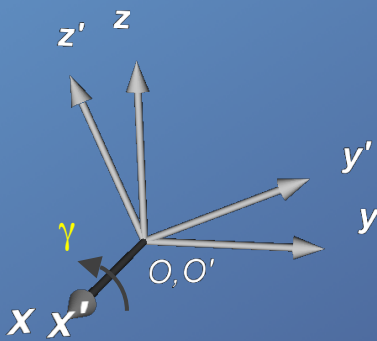
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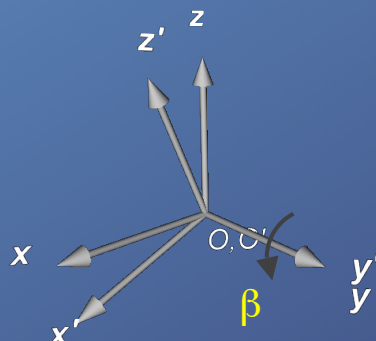
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The Rotation Matrix – Elementary Rotations

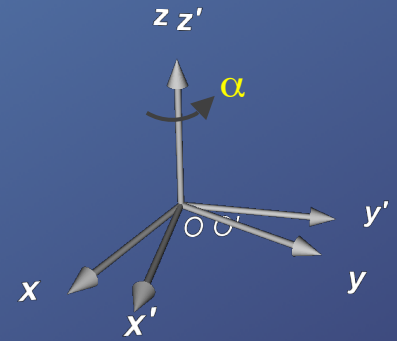
Elementary rotations are the rotations about the coordinate axis of the base frame; they are positive if they are made counter-clockwise.



$$R_x(\gamma) = \begin{bmatrix} 1 & 0 & 0 \\ 0 & c_\gamma & -s_\gamma \\ 0 & s_\gamma & c_\gamma \end{bmatrix}$$



$$R_y(\beta) = \begin{bmatrix} c_\beta & 0 & s_\beta \\ 0 & 1 & 0 \\ -s_\beta & 0 & c_\beta \end{bmatrix}$$



$$R_z(\alpha) = \begin{bmatrix} c_\alpha & -s_\alpha & 0 \\ s_\alpha & c_\alpha & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

For the elementary rotations:

$$R_k(-\vartheta) = R_k^T(\vartheta) \quad k = x, y, z$$



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The Rotation Matrix

A rotation matrix attains **three equivalent geometrical meanings**:

- It describes the mutual orientation between two coordinate frames; its column vectors are the direction cosines of the axes of the rotated frame with respect to the original frame (*as we just saw*).
- It represents the coordinate transformation between the coordinates of a point expressed in two different frames (with common origin).
- It is the operator that allows the rotation of a vector in the same coordinate frame.



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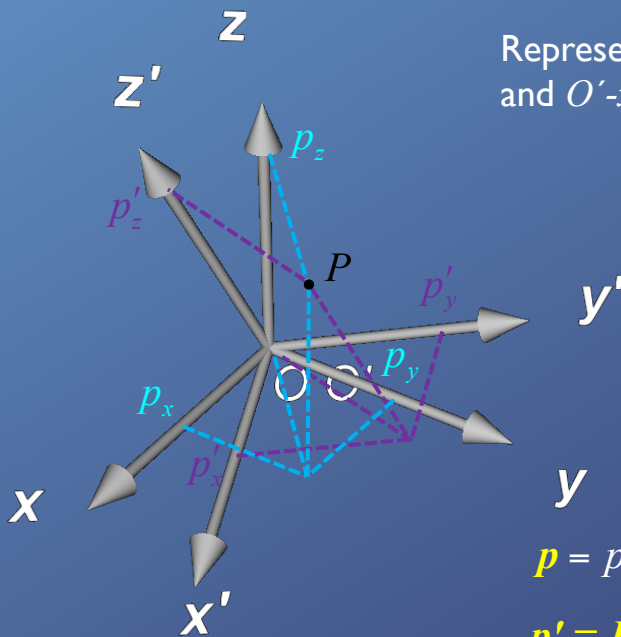
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The Rotation Matrix – Representation of a vector

The rotation matrix represents not only the orientation of a frame with respect to another frame, but it also describes the *transformation* of a vector from a frame to another frame with the same origin.



Representations of point P with respect to O - xyz and O' - $x'y'z'$ respectively:

$$p = \begin{bmatrix} p_x \\ p_y \\ p_z \end{bmatrix} \quad p' = \begin{bmatrix} p'_x \\ p'_y \\ p'_z \end{bmatrix}$$

These relate as follows:

$$p = p'_x x' + p'_y y' + p'_z z' = \begin{bmatrix} x' & y' & z' \end{bmatrix} \begin{bmatrix} p'_x \\ p'_y \\ p'_z \end{bmatrix} = R p'$$

$$p' = R^T p$$



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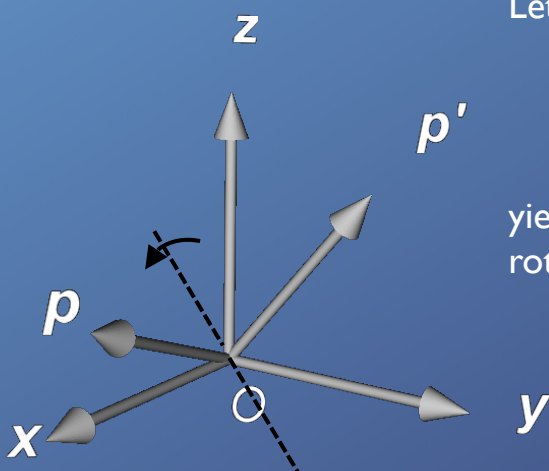
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The Rotation Matrix – Rotation of a vector

The rotation matrix can be also interpreted as the matrix operator allowing rotation of a vector by a given angle about an arbitrary axis in space.



Let p' be a vector expressed in O - xyz , then

$$p = R p'$$

yields a vector p with the same norm as that of p' but rotated with respect to p' according to the matrix R :

$$p^T p = p'^T R^T R p' = p'^T p'$$



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Composition of Rotation Matrices

Let p^0, p^1 and p^2 denote the expressions of vector p with respect to reference frames $O-x_0y_0z_0$, $O-x_1y_1z_1$ and $O-x_2y_2z_2$, respectively. Then, according to vector representation we have

$$\left. \begin{aligned} p^0 &= R_1^0 p^1 \\ p^1 &= R_2^1 p^2 \end{aligned} \right\} p^0 = R_1^0 R_2^1 p^2 = R_2^0 p^2$$

(R_i^j Rotation matrix of frame i with respect to frame j)

$R_2^0 = R_1^0 R_2^1 \rightarrow$ Composition of successive rotations is obtained by postmultiplication of the rotation matrices following the given order of rotations.



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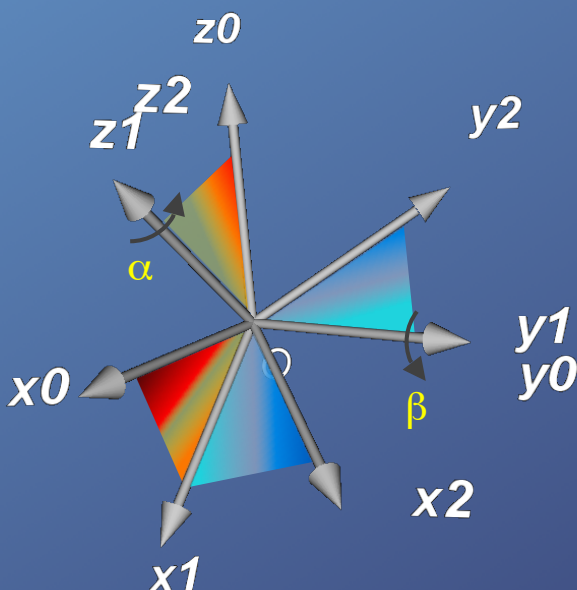
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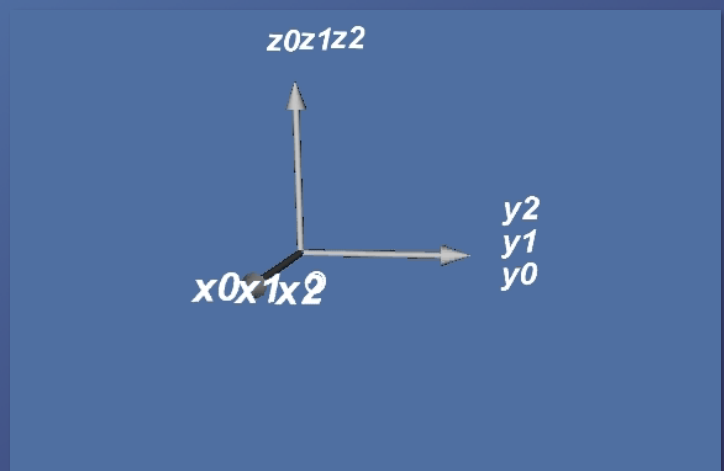
Composition of Rotation Matrices

The overall rotation matrix can be interpreted as a sequence of partial rotations, each expressed with respect to the previous one; the **current frame**.

Example: Rotate β degrees about y_0 followed by α degrees about z_1



$$R_2^0 = R_1^0 R_2^1 = R_y(\beta) R_z(\alpha)$$



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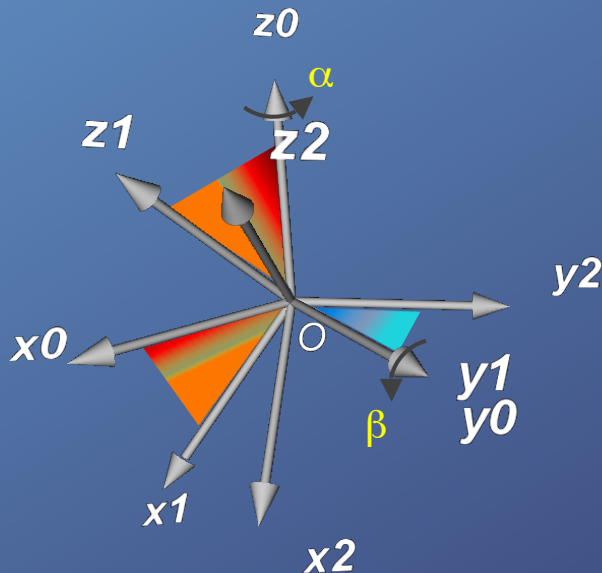
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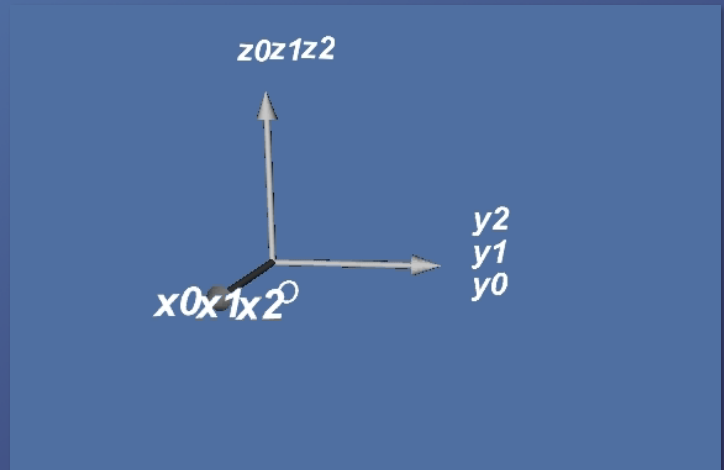
Composition of Rotation Matrices

Often it is desired to perform a sequence of rotations, each about the **fixed coordinate frame**, rather than the **current frame**.

Example: Rotate β degrees about y_0 followed by α degrees about z_0 instead of z_1 .



$$R_2^0 = R_1^0 R_0^1 R_2^1 R_1^0 = R_2^1 R_1^0 = R_z(\alpha) R_y(\beta)$$



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Composition of Rotation Matrices

Composition of successive rotations with respect to the **fixed frame** is obtained by **premultiplication** of the single rotation matrices in the order of the given sequence of rotations.

$$R_2^0 = R_2^1 R_1^0$$

Composition of successive rotations with respect to the **current frame** is obtained by **postmultiplication** of the single rotation matrices in the order of the given sequence of rotations.

$$R_2^0 = R_1^0 R_2^1$$

Matrix product does not commute, therefore, two rotations in general do not commute and its composition depends on the order of the single rotations

$$R_1^0 R_2^1 \neq R_2^1 R_1^0$$



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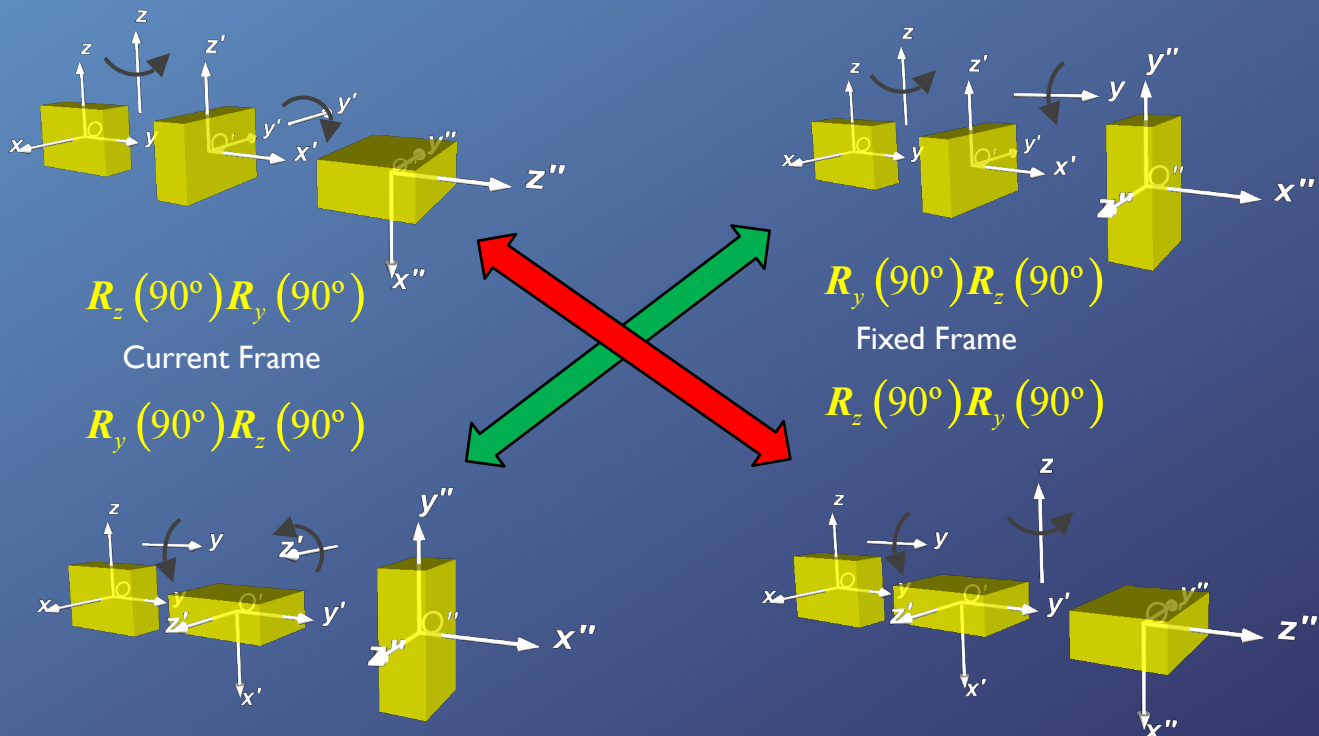
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Composition of Rotation Matrices

The order of rotations in the **fixed frame** commutes with respect to the order of rotations in the **current frame**:



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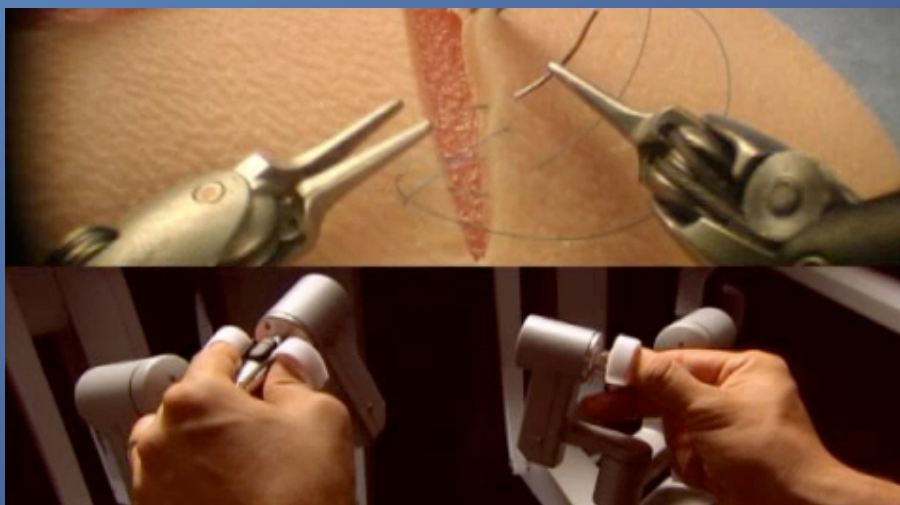
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The da Vinci Surgical System

<http://www.intuitivesurgical.com>



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Euler Angles

Rotation matrices give a redundant description of orientation – R possesses 9 elements which are not independent but related by the 6 constraints $R^T R = I$.

Therefore, a *minimal* representation of orientation needs only 3 parameters:

$$\phi = [\varphi \quad \vartheta \quad \psi]^T$$

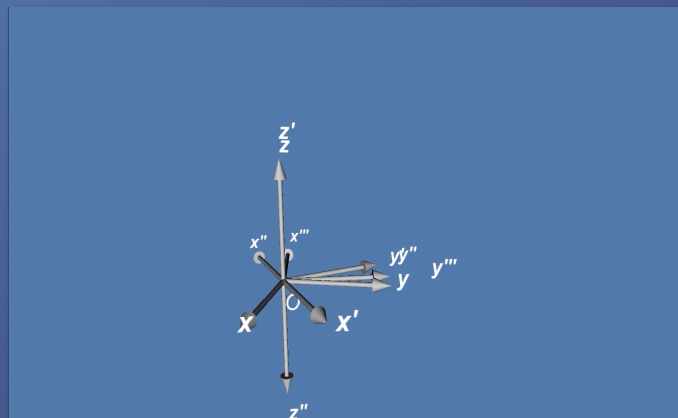
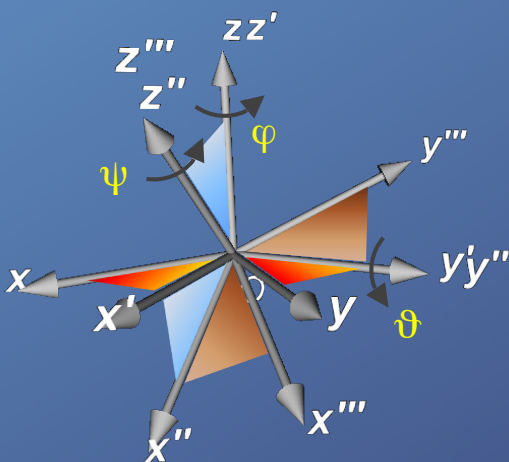
A generic rotation matrix may then be composed from a sequence of three elementary rotations, as long as two successive elementary rotations are not made along parallel axes. **Twelve** distinct sets of Euler angles result:

$$\begin{array}{cccc} XYZ & XZY & XYX & XZX \\ YZX & YXZ & YZY & YXY \\ ZXY & ZYX & ZXZ & ZYZ \end{array}$$

The most commonly used are the ZYZ and the ZYX (Roll – Pitch – Yaw)

Euler Angles – ZYZ set

$$R(\phi) = R_z(\varphi)R_y(\vartheta)R_z(\psi) = \begin{bmatrix} C_\varphi C_\vartheta C_\psi - S_\varphi S_\psi & -C_\varphi C_\vartheta S_\psi - S_\varphi C_\psi & C_\varphi S_\vartheta \\ S_\varphi C_\vartheta C_\psi + C_\varphi S_\psi & -S_\varphi C_\vartheta S_\psi + C_\varphi C_\psi & S_\varphi S_\vartheta \\ -S_\vartheta C_\psi & S_\vartheta S_\psi & C_\vartheta \end{bmatrix}$$



When $\vartheta = 0, \pi$, the successive rotations of φ and ψ are made about parallel axes, thus giving equivalent contributions to the rotation : **singularity**.

Euler Angles – ZYZ set

Solution of the inverse problem:

$$R = \begin{bmatrix} r_{11} & r_{12} & r_{13} \\ r_{21} & r_{22} & r_{23} \\ r_{31} & r_{32} & r_{33} \end{bmatrix} = \begin{bmatrix} c_\varphi c_\vartheta c_\psi - s_\varphi s_\psi & -c_\varphi c_\vartheta s_\psi - s_\varphi c_\psi & c_\varphi s_\vartheta \\ s_\varphi c_\vartheta c_\psi + c_\varphi s_\psi & -s_\varphi c_\vartheta s_\psi + c_\varphi c_\psi & s_\varphi s_\vartheta \\ -s_\vartheta c_\psi & s_\vartheta s_\psi & c_\vartheta \end{bmatrix}$$

$r_{13}, r_{23} \neq 0$

$$\vartheta \in]0, \pi[$$

$$\varphi = \text{Atan2}(r_{23}, r_{13})$$

$$\vartheta = \text{Atan2}\left(\sqrt{r_{13}^2 + r_{23}^2}, r_{33}\right) \quad \text{or}$$

$$\psi = \text{Atan2}(r_{32}, -r_{31})$$

$$\vartheta \in]-\pi, 0[$$

$$\varphi = \text{Atan2}(-r_{23}, -r_{13})$$

$$\vartheta = \text{Atan2}\left(-\sqrt{r_{13}^2 + r_{23}^2}, r_{33}\right)$$

$$\psi = \text{Atan2}(-r_{32}, r_{31})$$

($\rightarrow$ solve Problem 2.2 for the singularity)



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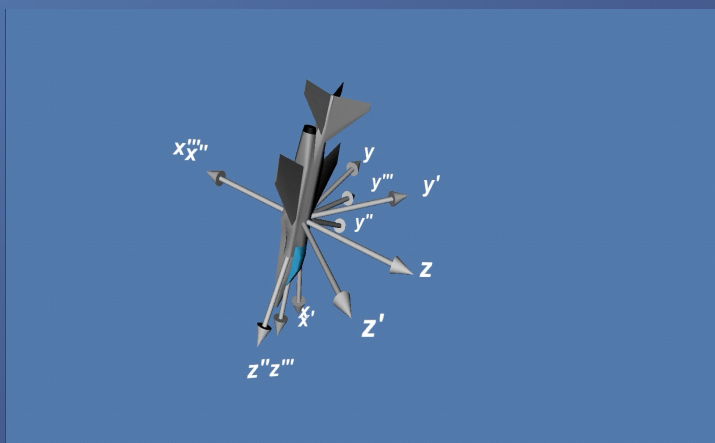
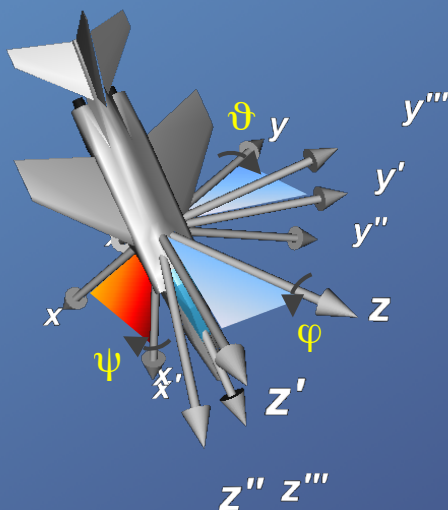
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Euler Angles – ZYX set (Roll – Pitch – Yaw)

$$R(\phi) = R_z(\varphi)R_y(\vartheta)R_x(\psi) = \begin{bmatrix} c_\varphi c_\vartheta & c_\varphi s_\vartheta s_\psi - s_\varphi c_\psi & c_\varphi s_\vartheta c_\psi + s_\varphi s_\psi \\ s_\varphi c_\vartheta & s_\varphi s_\vartheta s_\psi + c_\varphi c_\psi & s_\varphi s_\vartheta c_\psi + c_\varphi s_\psi \\ -s_\vartheta & c_\vartheta s_\psi & c_\vartheta c_\psi \end{bmatrix}$$



When $\vartheta = -\frac{\pi}{2}, \frac{\pi}{2}$, the successive rotations of φ and ψ are made about parallel axes, thus giving equivalent contributions to the rotation : **singularity**.



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Euler Angles – ZYX set (Roll – Pitch - Yaw)

Solution of the inverse problem:

$$R(\phi) = \begin{bmatrix} r_{11} & r_{12} & r_{13} \\ r_{21} & r_{22} & r_{23} \\ r_{31} & r_{32} & r_{33} \end{bmatrix} = \begin{bmatrix} c_\varphi c_\vartheta & c_\varphi s_\vartheta s_\psi - s_\varphi c_\psi & c_\varphi s_\vartheta c_\psi + s_\varphi s_\psi \\ s_\varphi c_\vartheta & s_\varphi s_\vartheta s_\psi + c_\varphi c_\psi & s_\varphi s_\vartheta c_\psi + c_\varphi s_\psi \\ -s_\vartheta & c_\vartheta s_\psi & c_\vartheta c_\psi \end{bmatrix}$$

$$\vartheta \in \left] -\frac{\pi}{2}, \frac{\pi}{2} \right[$$

$$\varphi = \text{Atan2}(r_{21}, r_{11})$$

$$\vartheta = \text{Atan2}\left(-r_{31}, \sqrt{r_{32}^2 + r_{33}^2}\right) \text{ or}$$

$$\psi = \text{Atan2}(r_{32}, r_{33})$$

$$\vartheta \in \left] \frac{\pi}{2}, \frac{3\pi}{2} \right[$$

$$\varphi = \text{Atan2}(-r_{21}, -r_{11})$$

$$\vartheta = \text{Atan2}\left(-r_{31}, -\sqrt{r_{32}^2 + r_{33}^2}\right)$$

$$\psi = \text{Atan2}(-r_{32}, -r_{33})$$

$$r_{11}, r_{21} \neq 0$$

($\rightarrow$ solve Problem 2.3 for the singularities)



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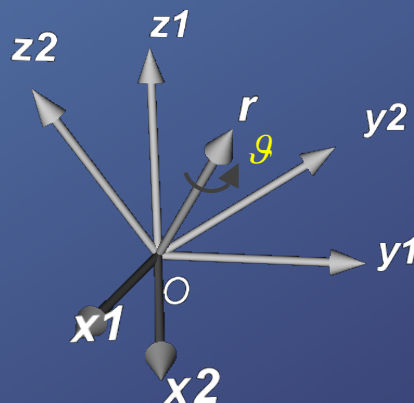
Angle and axis

A **nonminimal** representation of orientation is obtained, if it is expressed as a rotation of a given angle, ϑ , about a unit axis in space, $\mathbf{r}$, – **four parameters** are thus necessary.

This representation has clearer interpretation, especially for planning the orientation trajectory of a manipulator's end effector.

$$\mathbf{r} = \begin{bmatrix} r_x & r_y & r_z \end{bmatrix}^T, \quad \|\mathbf{r}\| = 1$$

$$\mathbf{R}_2^1 = R(\vartheta, \mathbf{r})$$



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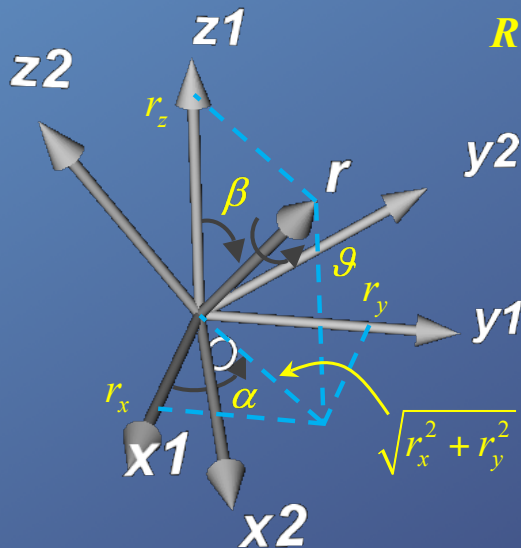
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Angle and axis

Derivation – rotate first $\mathbf{r}$ by the angles necessary to align it with axis $\mathbf{z}_1$, then rotate by ϑ about $\mathbf{z}_1$ and finally rotate by the angles necessary to align the unit vector with the initial direction:



$$R(\vartheta, \mathbf{r}) = R_z(\alpha) R_y(\beta) R_z(\vartheta) R_y(-\beta) R_z(-\alpha)$$

where

$$s_\alpha = \frac{r_y}{\sqrt{r_x^2 + r_y^2}}$$

$$c_\alpha = \frac{r_x}{\sqrt{r_x^2 + r_y^2}}$$

$$s_\beta = \sqrt{r_x^2 + r_y^2}$$

$$c_\beta = r_z$$



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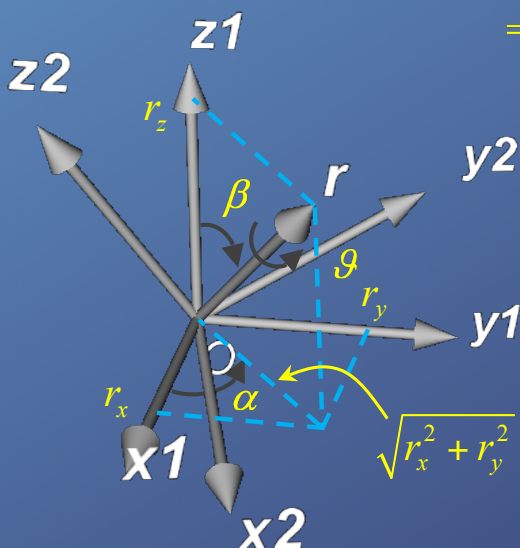
Angle and axis

Derivation – ... after substitution yields (→ solve Problem 2.4)

$$\begin{aligned} R(\vartheta, \mathbf{r}) &= R_z(\alpha) R_y(\beta) R_z(\vartheta) R_y(-\beta) R_z(-\alpha) \\ &= R_z(\alpha) R_y(\beta) R_z(\vartheta) [R_z(\alpha) R_y(\beta)]^T \end{aligned}$$

$$= \begin{bmatrix} r_x^2(1-c_\vartheta) + c_\vartheta & r_x r_y(1-c_\vartheta) - r_z s_\vartheta & r_x r_z(1-c_\vartheta) + r_y s_\vartheta \\ r_x r_y(1-c_\vartheta) + r_z s_\vartheta & r_y^2(1-c_\vartheta) + c_\vartheta & r_y r_z(1-c_\vartheta) - r_x s_\vartheta \\ r_x r_z(1-c_\vartheta) - r_y s_\vartheta & r_y r_z(1-c_\vartheta) + r_x s_\vartheta & r_z^2(1-c_\vartheta) + c_\vartheta \end{bmatrix}$$

$$= I_3 + s_\vartheta \mathbf{S}(\mathbf{r}) + (1-c_\vartheta) \mathbf{S}(\mathbf{r}) \mathbf{S}(\mathbf{r}), \quad \mathbf{S}(\mathbf{r}) = \begin{bmatrix} 0 & -r_z & r_y \\ r_z & 0 & -r_x \\ -r_y & r_x & 0 \end{bmatrix}$$



Notice that

$$R(\vartheta, \mathbf{r}) = R(-\vartheta, -\mathbf{r})$$

A rotation by $-\vartheta$ about $-\mathbf{r}$ cannot be distinguished from a rotation by ϑ about $\mathbf{r} \Rightarrow$ representation is not unique.



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Angle and axis

Solution of the inverse problem – compute the angle and axis corresponding to a given rotation matrix:

$$R(\vartheta, \mathbf{r}) = \begin{bmatrix} r_{11} & r_{12} & r_{13} \\ r_{21} & r_{22} & r_{23} \\ r_{31} & r_{32} & r_{33} \end{bmatrix} = \begin{bmatrix} r_x^2(1-c_\vartheta) + c_\vartheta & r_x r_y(1-c_\vartheta) - r_z s_\vartheta & r_x r_z(1-c_\vartheta) + r_y s_\vartheta \\ r_x r_y(1-c_\vartheta) + r_z s_\vartheta & r_y^2(1-c_\vartheta) + c_\vartheta & r_y r_z(1-c_\vartheta) - r_x s_\vartheta \\ r_x r_z(1-c_\vartheta) - r_y s_\vartheta & r_y r_z(1-c_\vartheta) + r_x s_\vartheta & r_z^2(1-c_\vartheta) + c_\vartheta \end{bmatrix}$$

From the diagonal elements: $\vartheta = \cos^{-1} \left(\frac{r_{11} + r_{22} + r_{33} - 1}{2} \right)$

From the off diagonal elements: $\mathbf{r} = \frac{1}{2s_\vartheta} \begin{bmatrix} r_{32} - r_{23} \\ r_{13} - r_{31} \\ r_{21} - r_{12} \end{bmatrix}, \quad \text{for } s_\vartheta \neq 0$

For $\vartheta = 0$, $\mathbf{r}$ becomes arbitrary $\Rightarrow$ singularity

For $\vartheta = \pi$, we can have $\mathbf{r}$ or $-\mathbf{r}$ ($\rightarrow$ solve Problem 2.5)



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Angle and axis

Solution of the inverse problem – compute the angle and axis corresponding to a given rotation matrix:

If $r_{11} + r_{22} + r_{33} = 3$

$$\vartheta = 0,$$

$$R(0, \mathbf{r}) = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$\mathbf{r}$ is arbitrary

If $r_{11} + r_{22} + r_{33} = -1$

$$\vartheta = \pi,$$

$$R(\pi, \pm \mathbf{r}) = \begin{bmatrix} 2r_x^2 - 1 & 2r_x r_y & 2r_x r_z \\ 2r_x r_y & 2r_y^2 - 1 & 2r_y r_z \\ 2r_x r_z & 2r_y r_z & 2r_z^2 - 1 \end{bmatrix}$$

$\pm \mathbf{r}$ may be taken from any one column of $\mathbf{R}$

For best numerical conditioning choose the column with $\max(\text{diag}(\mathbf{R}))$

For example, for the first column:

$$r_x = \pm \sqrt{\frac{r_{11} + 1}{2}} \quad r_y = \frac{r_{21}}{2r_x} \quad r_z = \frac{r_{31}}{2r_x}$$



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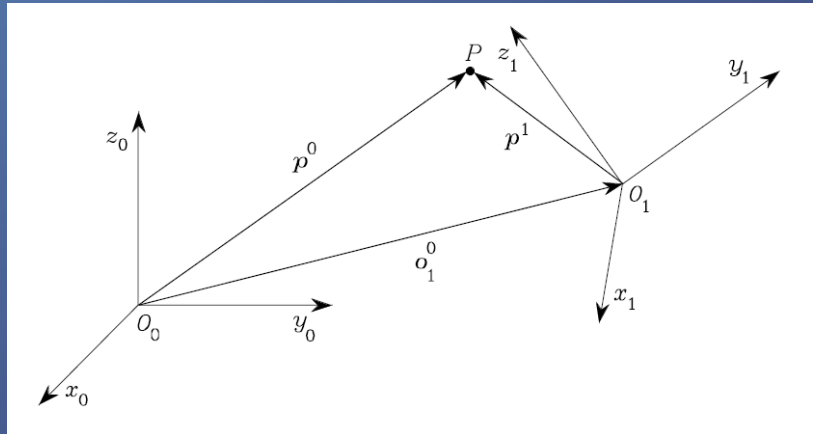
Homogeneous transformation

A **compact** representation for the **position** and **orientation** of a rigid body – It allows for the coordinate transformation (**translation + rotation**) of a bound vector between two frames.

$$p^0 = o_1^0 + R_1^0 p^1$$

furthermore:

$$\begin{aligned} p^1 &= -R_1^{0T} o_1^0 + R_1^{0T} p^0 \\ &= -R_0^1 o_1^0 + R_0^1 p^0 \end{aligned}$$



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Homogeneous transformation

The **homogeneous representation** of a generic vector p can be introduced as the vector $\tilde{p}$ formed by adding a fourth **unit** component

$$\tilde{p} = \begin{bmatrix} p \\ 1 \end{bmatrix}$$

$$A_1^0 = \begin{bmatrix} R_1^0 & o_1^0 \\ 0^T & 1 \end{bmatrix}$$

Homogeneous transformation matrix

$$p^0 = o_1^0 + R_1^0 p^1$$



$$\tilde{p}^0 = A_1^0 \tilde{p}^1$$

In general:

$$\tilde{p}^0 = A_1^0 A_2^1 \dots A_n^{n-1} \tilde{p}^n$$



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Homogeneous transformation

The inverse transformation can be expressed as:

$$\tilde{p}^1 = A_0^1 \tilde{p}^0 = (A_1^0)^{-1} \tilde{p}^0$$

where:

$$A_0^1 = \begin{bmatrix} R_1^{0T} & -R_1^{0T} o_1^0 \\ 0^T & 1 \end{bmatrix} = \begin{bmatrix} R_0^1 & -R_0^1 o_1^0 \\ 0^T & 1 \end{bmatrix}$$

$$A^{-1} \neq A^T \quad (\rightarrow \text{solve Problem 2.10})$$



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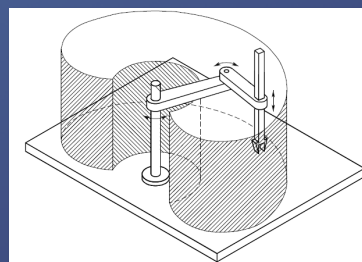
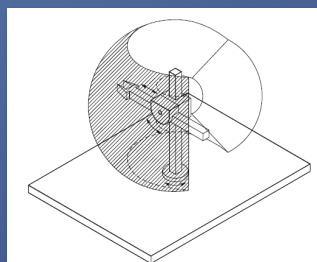
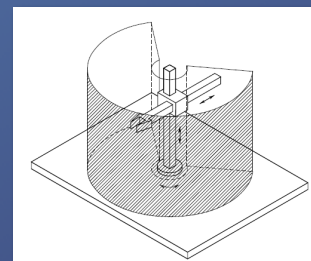
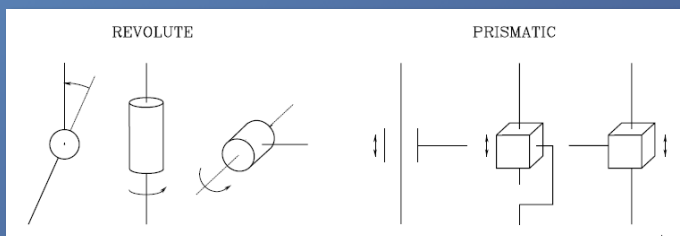
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Robótica de Manipulação

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Direct Kinematics

The structure of a manipulator forms a **kinematic chain** composed of links and joints. One end of the chain is constrained to a base, and an end-effector (gripper, tool) is connected to the other.



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Robótica de Manipulação

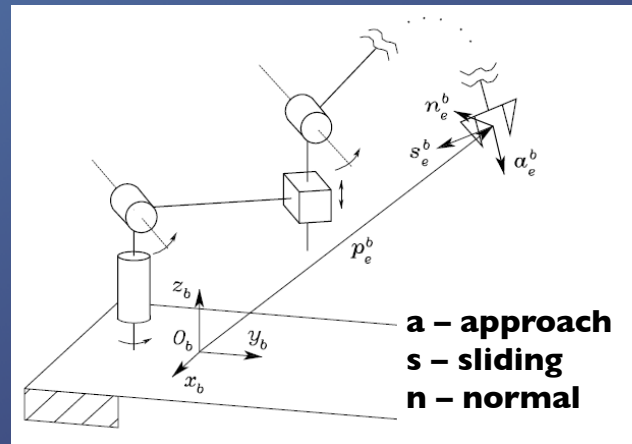
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Direct Kinematics

The mechanical structure of a manipulator is characterized by a number of degrees of freedom (DOFs) that uniquely determine its **posture** (the **pose of all the rigid bodies composing the chain**). Each DOF is typically associated with a joint articulation and constitutes a joint variable.

$$T_e^b(q) = \begin{bmatrix} n_e^b(q) & s_e^b(q) & a_e^b(q) & p_e^b(q) \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

Where q is the $(n \times 1)$ vector of joint variables



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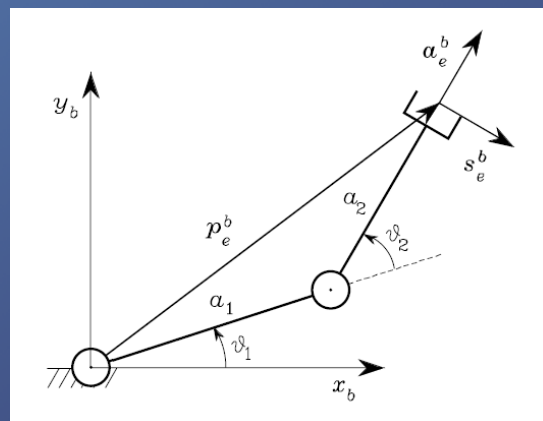
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Direct Kinematics

Simple example: determine the homogeneous transformation matrix

$$T_e^b(q) = \begin{bmatrix} n_e^b & s_e^b & a_e^b & p_e^b \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

$$= \begin{bmatrix} 0 & s_{12} & c_{12} & a_1 c_1 + a_2 c_{12} \\ 0 & -c_{12} & s_{12} & a_1 s_1 + a_2 s_{12} \\ 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$



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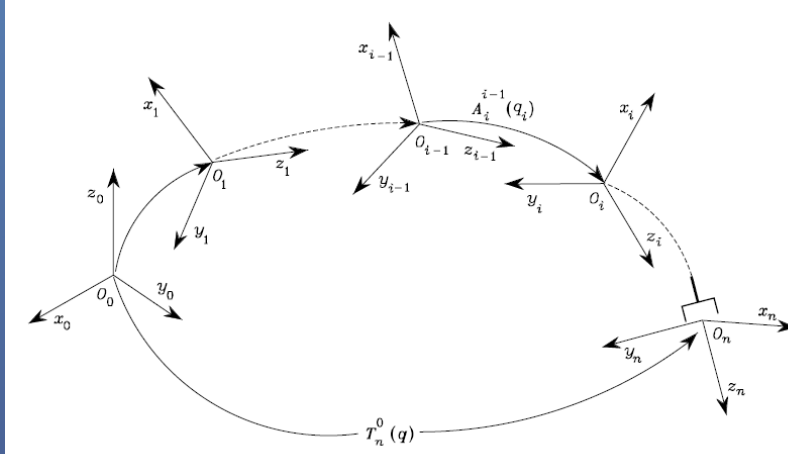
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Robótica de Manipulação

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Open Chain

The computation of the **direct kinematics function** is recursive and is obtained by simple products of the homogeneous transformation matrices



$$T_n^0(q) = A_1^0(q_1) A_2^1(q_2) \dots A_n^{n-1}(q_n)$$

$$T_e^b(q) = T_0^b T_n^0(q) T_e^n$$



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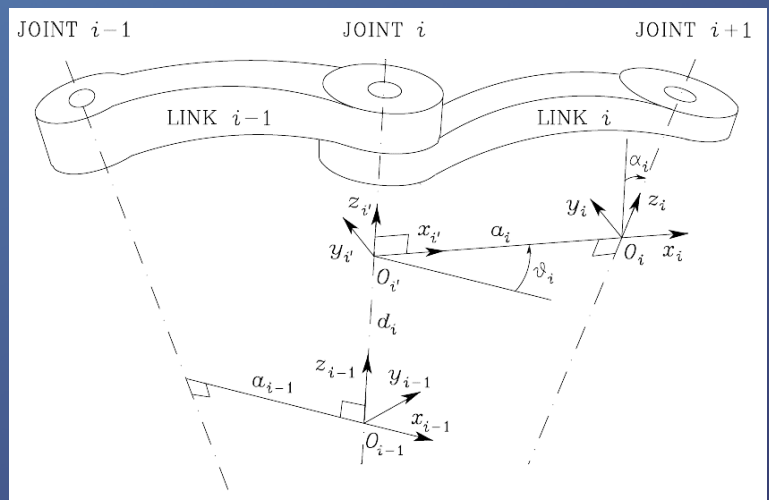
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Denavit-Hartenberg (DH) Convention

A systematic and general **method to define the relative position and orientation of two consecutive links, in order to compute the direct kinematics equation** for an open-chain manipulator.

- 1) Choose axis z_i along the axis of Joint $i + 1$
- 2) Locate the origin O_i at the intersection of axis z_i with the common normal to axes z_{i-1} and z_i . Also, locate $O_{i'}$ at the intersection of the common normal with axis z_{i-1} .
- 3) Choose axis x_i along the common normal to axes z_{i-1} and z_i with direction from Joint i to Joint $i + 1$.
- 4) Choose axis y_i so as to complete a right-handed frame.



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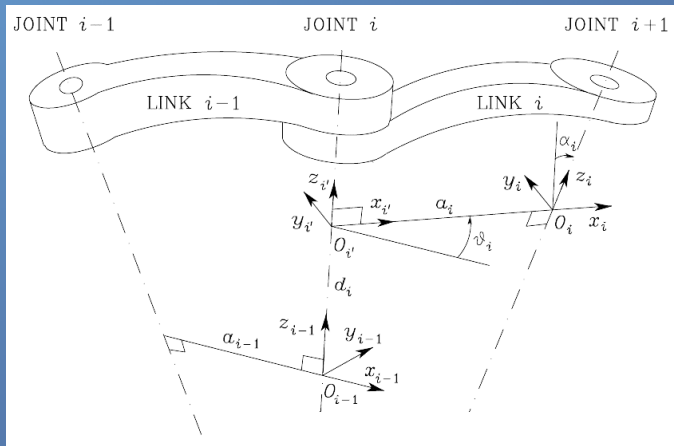
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Robótica de Manipulação

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Denavit-Hartenberg (DH) Convention



$$A_i^{i'} = \begin{bmatrix} 1 & 0 & 0 & a_i \\ 0 & c_{\alpha_i} & -s_{\alpha_i} & 0 \\ 0 & s_{\alpha_i} & c_{\alpha_i} & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

$$A_i^{i-1} = \begin{bmatrix} c_{\vartheta_i} & -s_{\vartheta_i} & 0 & 0 \\ s_{\vartheta_i} & c_{\vartheta_i} & 0 & 0 \\ 0 & 0 & 1 & d_i \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

$$A_i^{i-1}(q_i) = A_i^{i-1} A_i^{i'} = \begin{bmatrix} c_{\vartheta_i} & -s_{\vartheta_i} c_{\alpha_i} & s_{\vartheta_i} s_{\alpha_i} & a_i c_{\vartheta_i} \\ s_{\vartheta_i} & c_{\vartheta_i} c_{\alpha_i} & -c_{\vartheta_i} s_{\alpha_i} & a_i s_{\vartheta_i} \\ 0 & s_{\alpha_i} & c_{\alpha_i} & d_i \\ 0 & 0 & 0 & 1 \end{bmatrix}$$



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Robótica de Manipulação

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Denavit-Hartenberg (DH) Convention

The DH convention gives a nonunique definition of the link frame in the following cases:

- For **Frame 0**, only the direction of axis z_0 is specified; then O_0 and x_0 can be arbitrarily chosen.
- For **Frame n**, since there is no Joint $n + 1$, z_n is not uniquely defined while x_n has to be normal to axis z_{n-1} . Typically, Joint n is revolute, and thus z_n is to be aligned with the direction of z_{n-1} .
- When two consecutive axes are parallel, the common normal between them is not uniquely defined.
- When two consecutive axes intersect, the direction of x_i is arbitrary.
- When Joint i is prismatic, the direction of z_{n-1} is arbitrary.



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Robótica de Manipulação

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Denavit-Hartenberg (DH) Convention

The DH parameters:

Once the link frames have been established, the position and orientation of Frame i with respect to Frame $i-1$ are completely specified by the following parameters:

a_i distance between O_i and O_p ,

d_i coordinate of O_p along z_{i-1}

α_i angle between axes z_{i-1} and z_i about axis x_i to be taken positive when rotation is made counter-clockwise

ϑ_i angle between axes x_{i-1} and x_i about axis z_{i-1} to be taken positive when rotation is made counter-clockwise

a_i and α_i are constant. Of the remaining two parameters, only one is variable depending on the type of joint:

- if Joint i is revolute the variable is ϑ_i ,
- if Joint i is prismatic the variable is d_i .



Denavit-Hartenberg (DH) Convention

Expressing the homogeneous transformation:

- Choose a frame aligned with Frame $i-1$.
- Translate the chosen frame by d_i along axis z_{i-1} and rotate it by ϑ_i about axis z_{i-1} ; this sequence aligns the current frame with Frame i and is described by the homogeneous transformation matrix

$$A_{i'}^{i-1} = \begin{bmatrix} c_{\vartheta_i} & -s_{\vartheta_i} & 0 & 0 \\ s_{\vartheta_i} & c_{\vartheta_i} & 0 & 0 \\ 0 & 0 & 1 & d_i \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

- Translate the frame aligned with Frame i by a_i along axis x_i and rotate it by α_i about axis x_i ; this sequence aligns the current frame with Frame i and is described by the homogeneous transformation matrix

$$A_i^{i'} = \begin{bmatrix} 1 & 0 & 0 & a_i \\ 0 & c_{\alpha_i} & -s_{\alpha_i} & 0 \\ 0 & s_{\alpha_i} & c_{\alpha_i} & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$



Denavit-Hartenberg (DH) Convention

Expressing the homogeneous transformation:

- The resulting coordinate transformation is obtained by postmultiplication of the single transformations as

$$A_i^{i-1}(q_i) = A_i^{i-1} A_i^{i'} = \begin{bmatrix} c\vartheta_i & -s\vartheta_i c\alpha_i & s\vartheta_i s\alpha_i & a_i c\vartheta_i \\ s\vartheta_i & c\vartheta_i c\alpha_i & -c\vartheta_i s\alpha_i & a_i s\vartheta_i \\ 0 & s\alpha_i & c\alpha_i & d_i \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

- Which is a function only of the joint variable q_i , meaning:

ϑ_i for a revolute joint

d_i for a prismatic joint



Denavit-Hartenberg (DH) Convention

Algorithmic form:

1. Find and number consecutively the joint axes; set the directions of axes $z_0, \dots, z_{n-1}$.
2. Choose **Frame 0** by locating the origin on axis z_0 ; axes x_0 and y_0 are chosen so as to obtain a right-handed frame. If feasible, it is worth choosing **Frame 0** to coincide with the base frame.

Execute steps from 3 to 5 for $i = 1, \dots, n - 1$:

3. Locate the origin O_i at the intersection of z_i with the common normal to axes z_{i-1} and z_i . If axes z_{i-1} and z_i are parallel and **Joint i** is **revolute**, then locate O_i so that $d_i = 0$; if **Joint i** is **prismatic**, locate O_i at a reference position for the joint range, e.g., a mechanical limit.
4. Choose axis x_i along the common normal to axes z_{i-1} and z_i with direction from **Joint i** to **Joint i + 1**.
5. Choose axis y_i so as to obtain a right-handed frame.



Denavit-Hartenberg (DH) Convention

Algorithmic form:

To complete:

6. Choose **Frame n** ; if **Joint n** is **revolute**, then align $\mathbf{z}_n$ with $\mathbf{z}_{n-1}$, otherwise, if **Joint n** is **prismatic**, then choose $\mathbf{z}_n$ arbitrarily. Axis $\mathbf{x}_n$ is set according to step 4.
7. For $i = 1, \dots, n$, form the table of parameters $\mathbf{a}_i, \mathbf{d}_i, \alpha_i, \vartheta_i$.
8. On the basis of the parameters in 7, compute the homogeneous transformation matrices $A_i^{i-1}(q_i)$ for $i = 1, \dots, n$.
9. Compute the homogeneous transformation $T_n^0(q) = A_1^0 \dots A_n^{n-1}$ that yields the position and orientation of **Frame n** with respect to **Frame 0**.
10. Given T_0^b and T_e^n , compute the direct kinematics function as $T_e^b(q) = T_0^b T_n^0(q) T_e^n$ that yields the position and orientation of the end-effector frame with respect to the base frame.



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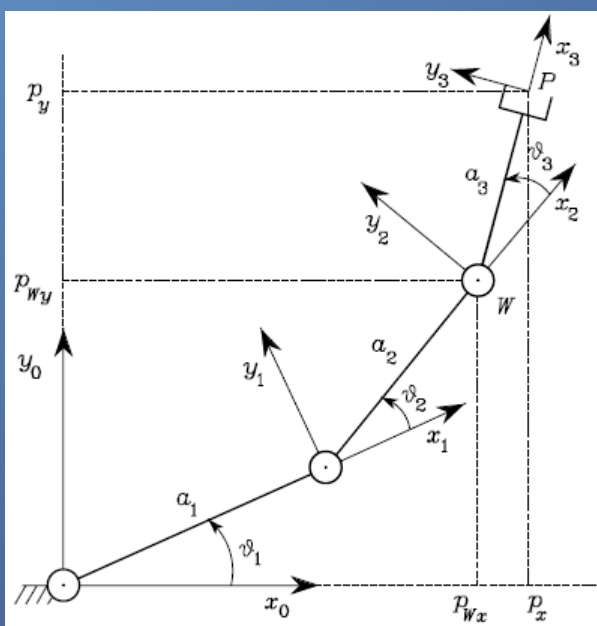
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Robótica de Manipulação

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Denavit-Hartenberg (DH) Convention

Example: Three-link Planar Arm



DH parameters

Link	a_i	α_i	d_i	ϑ_i
1	a_1	0	0	ϑ_1
2	a_2	0	0	ϑ_2
3	a_3	0	0	ϑ_3

Homogeneous transformation matrix of joint i

$$A_i^{i-1}(\vartheta_i) = \begin{bmatrix} c_i & -s_i & 0 & a_i c_i \\ s_i & c_i & 0 & a_i s_i \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \quad i = 1, 2, 3.$$

Direct Kinematics Function

$$T_3^0(q) = A_1^0 A_2^1 A_3^2 =$$

$$= \begin{bmatrix} c_{123} & -s_{123} & 0 & a_1 c_1 + a_2 c_{12} + a_3 c_{123} \\ s_{123} & c_{123} & 0 & a_1 s_1 + a_2 s_{12} + a_3 s_{123} \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$



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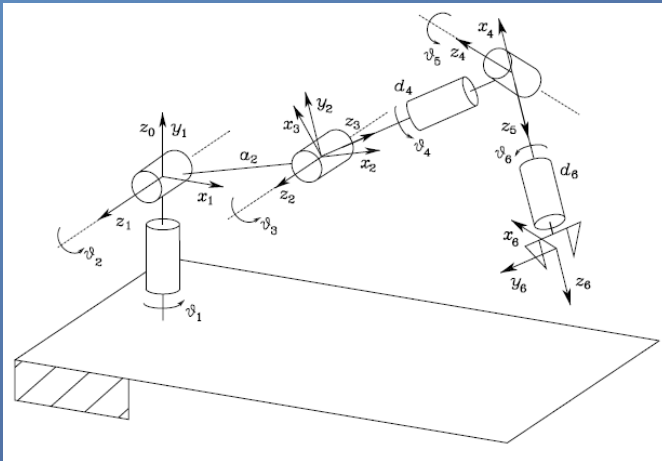
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Robótica de Manipulação

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Denavit-Hartenberg (DH) Convention

Example: Anthropomorphic Arm with Spherical Wrist



DH parameters

Link	a_i	α_i	d_i	θ_i
1	0	$\pi/2$	0	θ_1
2	a_2	0	0	θ_2
3	0	$\pi/2$	0	θ_3
4	0	$-\pi/2$	d_4	θ_4
5	0	$\pi/2$	0	θ_5
6	0	0	d_6	θ_6

Resulting position and orientation (components of Direct Kinematics Function)

$$\begin{aligned}
 p_6^0 &= \begin{bmatrix} a_2 c_1 c_2 + d_4 c_1 s_{23} + d_6 (c_1 (c_{23} c_4 s_5 + s_{23} c_5) + s_1 s_4 s_5) \\ a_2 s_1 c_2 + d_4 s_1 s_{23} + d_6 (s_1 (c_{23} c_4 s_5 + s_{23} c_5) - c_1 s_4 s_5) \\ a_2 s_2 - d_4 c_{23} + d_6 (s_{23} c_4 s_5 - c_{23} c_5) \end{bmatrix} & s_6^0 &= \begin{bmatrix} c_1 (-c_{23} (c_4 c_5 s_6 + s_4 c_6) + s_{23} s_5 s_6) + s_1 (-s_4 c_5 s_6 + c_4 c_6) \\ s_1 (-c_{23} (c_4 c_5 s_6 + s_4 c_6) + s_{23} s_5 s_6) - c_1 (-s_4 c_5 s_6 + c_4 c_6) \\ -s_{23} (c_4 c_5 s_6 + s_4 c_6) - c_{23} s_5 s_6 \end{bmatrix} \\
 n_6^0 &= \begin{bmatrix} c_1 (c_{23} (c_4 c_5 c_6 - s_4 s_6) - s_{23} s_5 c_6) + s_1 (s_4 c_5 c_6 + c_4 s_6) \\ s_1 (c_{23} (c_4 c_5 c_6 - s_4 s_6) - s_{23} s_5 c_6) - c_1 (s_4 c_5 c_6 + c_4 s_6) \\ s_{23} (c_4 c_5 c_6 - s_4 s_6) + c_{23} s_5 c_6 \end{bmatrix} & a_6^0 &= \begin{bmatrix} c_1 (c_{23} c_4 s_5 + s_{23} c_5) + s_1 s_4 s_5 \\ s_1 (c_{23} c_4 s_5 + s_{23} c_5) - c_1 s_4 s_5 \\ s_{23} c_4 s_5 - c_{23} c_5 \end{bmatrix}
 \end{aligned}$$



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Inverse Kinematics

The inverse kinematics problem consists of the determination of the joint variables corresponding to a given end-effector position and orientation.

It is more complex than the Direct Kinematics problem:

- The equations to solve are in general **nonlinear**, and thus it is not always possible to find a closed-form solution.
- **Multiple solutions** may exist.
- **Infinite solutions** may exist, e.g., in the case of a kinematically redundant manipulator.
- There might be **no admissible solutions**, in view of the manipulator kinematic structure.



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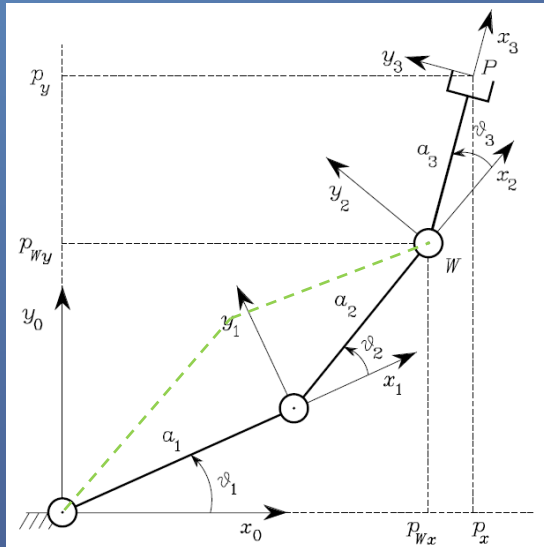
Robótica de Manipulação

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Inverse Kinematics

Solution of the three-link planar arm: $p_x, p_y, \phi \rightarrow \vartheta_1, \vartheta_2, \vartheta_3$

$$T_3^0(q) = \begin{bmatrix} c_{123} & -s_{123} & 0 & a_1 c_1 + a_2 c_{12} + a_3 c_{123} \\ s_{123} & c_{123} & 0 & a_1 s_1 + a_2 s_{12} + a_3 s_{123} \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$



$$\phi = \vartheta_1 + \vartheta_2 + \vartheta_3$$

$$p_{Wx} = p_x - a_3 c_\phi = a_1 c_1 + a_2 c_{12}$$

$$p_{Wy} = p_y - a_3 s_\phi = a_1 s_1 + a_2 s_{12}$$

$$p_{Wx}^2 + p_{Wy}^2 = a_1^2 + a_2^2 + 2a_1 a_2 c_2$$

$$c_2 = \frac{p_{Wx}^2 + p_{Wy}^2 - a_1^2 - a_2^2}{2a_1 a_2} \quad s_2 = \pm \sqrt{1 - c_2^2}$$

Law of Cosines

$$\gamma = \arccos\left(\frac{a^2 + b^2 - c^2}{2ab}\right)$$

Solution exists if: $-1 \leq c_2 \leq 1$

$$\vartheta_2 = \text{Atan2}(s_2, c_2)$$



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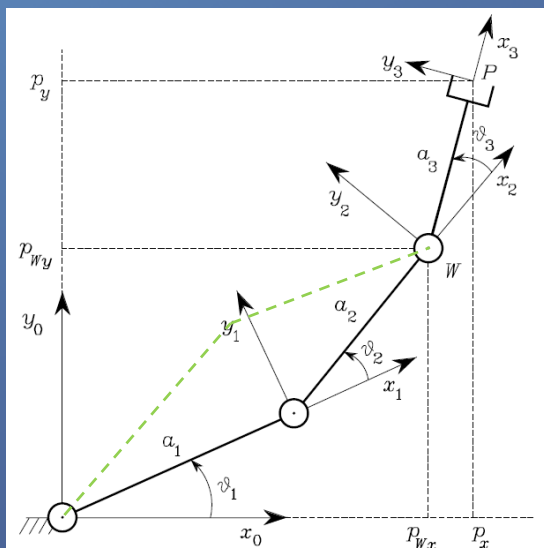
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Inverse Kinematics

Solution of the three-link planar arm: $p_x, p_y, \phi \rightarrow \vartheta_1, \vartheta_2, \vartheta_3$

$$T_3^0(q) = \begin{bmatrix} c_{123} & -s_{123} & 0 & a_1 c_1 + a_2 c_{12} + a_3 c_{123} \\ s_{123} & c_{123} & 0 & a_1 s_1 + a_2 s_{12} + a_3 s_{123} \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$



Substituting ϑ_2 in

$$p_{Wx} = p_x - a_3 c_\phi = a_1 c_1 + a_2 c_{12}$$

$$p_{Wy} = p_y - a_3 s_\phi = a_1 s_1 + a_2 s_{12}$$

results:

$$c_{12} = c_1 c_2 - s_1 s_2$$

$$s_{12} = s_1 c_2 + c_1 s_2$$

$$s_1 = \frac{(a_1 + a_2 c_2) p_{Wy} - a_2 s_2 p_{Wx}}{p_{Wx}^2 + p_{Wy}^2}$$

$$c_1 = \frac{(a_1 + a_2 c_2) p_{Wx} + a_2 s_2 p_{Wy}}{p_{Wx}^2 + p_{Wy}^2}$$

and finally:

$$\vartheta_1 = \text{Atan2}(s_1, c_1)$$

$$\vartheta_3 = \phi - \vartheta_1 - \vartheta_2$$



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Robótica de Manipulação

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Inverse Kinematics

Solution of Manipulators with Spherical Wrist

A 6-DOF kinematic structure has closed-form inverse kinematics solutions if:

- three consecutive revolute joint axes intersect at a common point, like for the spherical wrist;
- three consecutive revolute joint axes are parallel.

Algebraic or geometric intuition is required to obtain closed-form solutions.

This allows us to address the inverse kinematics problem by dividing it into two sub-problems, since **the solution for the position is decoupled from that for the orientation**.



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Robótica de Manipulação

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Inverse Kinematics

Solution of Manipulators with Spherical Wrist

In the case of a (non-redundant) three-DOF arm, the inverse kinematics can be solved according to the following steps:

- Compute the wrist position $\mathbf{p}_W(q_1, q_2, q_3)$
- Solve inverse kinematics for (q_1, q_2, q_3) .
- Compute $R_3^0(q_1, q_2, q_3)$.
- Compute $R_6^3(\vartheta_4, \vartheta_5, \vartheta_6) = (R_3^0)^T R$.
- Solve inverse kinematics for orientation $(\vartheta_4, \vartheta_5, \vartheta_6)$.



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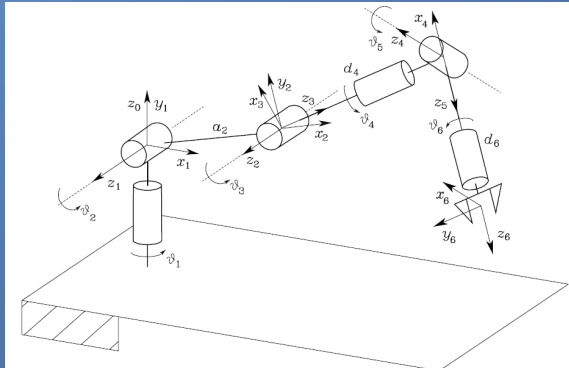
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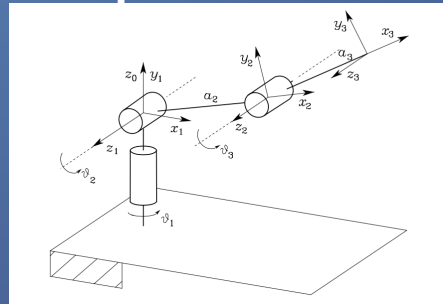
Inverse Kinematics

Solution of the Anthropomorphic arm with spherical wrist:



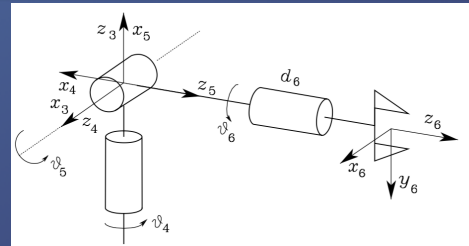
Link	a_i	α_i	d_i	ϑ_i
1	0	$\pi/2$	0	ϑ_1
2	a_2	0	0	ϑ_2
3	0	$\pi/2$	0	ϑ_3
4	0	$-\pi/2$	d_4	ϑ_4
5	0	$\pi/2$	0	ϑ_5
6	0	0	d_6	ϑ_6

Arm



Link	a_i	α_i	d_i	ϑ_i
1	0	$\pi/2$	0	ϑ_1
2	a_2	0	0	ϑ_2
3	a_3	0	0	ϑ_3

Wrist



Link	a_i	α_i	d_i	ϑ_i
4	0	$-\pi/2$	0	ϑ_4
5	0	$\pi/2$	0	ϑ_5
6	0	0	d_6	ϑ_6



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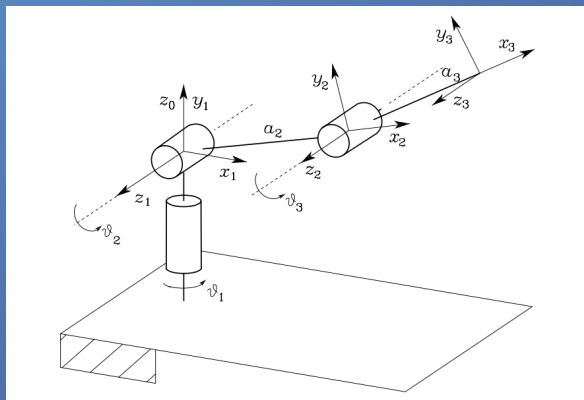
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Robótica de Manipulação

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Inverse Kinematics

Solution of the Anthropomorphic arm with spherical wrist:



Solution exists if:

$$-1 \leq c_3 \leq 1$$

1 – Calculate p_W $p_W = p_e - d_6 a_e$

$$p_W x = c_1(a_2 c_2 + a_3 c_{23})$$

$$p_W y = s_1(a_2 c_2 + a_3 c_{23})$$

$$p_W z = a_2 s_2 + a_3 s_{23}.$$

2 – Solve for ϑ_3

$$p_W^2 x + p_W^2 y + p_W^2 z = a_2^2 + a_3^2 + 2a_2 a_3 c_3$$

$$c_3 = \frac{p_W^2 x + p_W^2 y + p_W^2 z - a_2^2 - a_3^2}{2a_2 a_3}$$

$$s_3 = \pm \sqrt{1 - c_3^2} \quad \text{Two solutions}$$

$$\vartheta_3 = \text{Atan2}(s_3, c_3)$$



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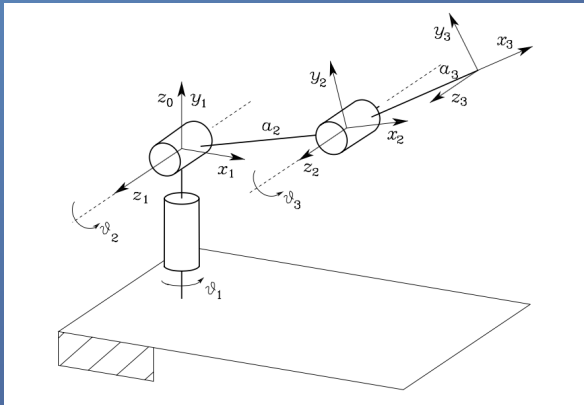
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Robótica de Manipulação

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Inverse Kinematics

Solution of the Anthropomorphic arm with spherical wrist:



3 – Solve for ϑ_2

$$p_{Wx}^2 + p_{Wy}^2 = (a_2 c_2 + a_3 c_2 c_3)^2$$

$$p_{Wz} = a_2 s_2 + a_3 s_2 c_3$$

$$c_2 = \frac{\pm \sqrt{p_{Wx}^2 + p_{Wy}^2} (a_2 + a_3 c_3) + p_{Wz} a_3 c_3}{a_2^2 + a_3^2 + 2a_2 a_3 c_3}$$

$$s_2 = \frac{p_{Wz} (a_2 + a_3 c_3) \mp \sqrt{p_{Wx}^2 + p_{Wy}^2} a_3 c_3}{a_2^2 + a_3^2 + 2a_2 a_3 c_3}.$$

Four solutions

$$\vartheta_2 = \text{Atan2}(s_2, c_2)$$



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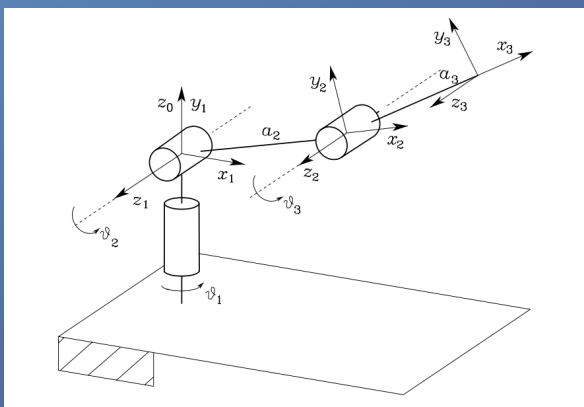
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Inverse Kinematics

Solution of the Anthropomorphic arm with spherical wrist:



4 – Solve for ϑ_1

$$\vartheta_{1,I} = \text{Atan2}(p_{Wy}, p_{Wx})$$

$$\vartheta_{1,II} = \text{Atan2}(-p_{Wy}, -p_{Wx}).$$

Two solutions



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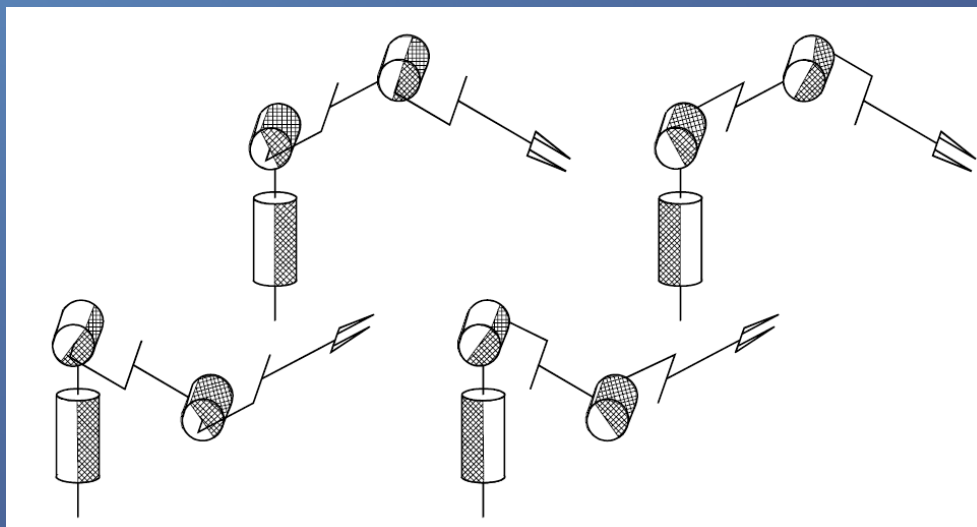
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Inverse Kinematics

Solution of the Anthropomorphic arm with spherical wrist:

Possible configurations of the arm



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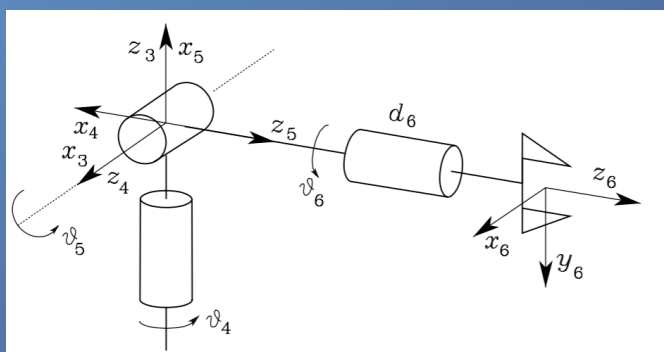
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Inverse Kinematics

Solution of the Anthropomorphic arm with spherical wrist:



5 – Calculate R_6^3

$$R_3^0(q_1, q_2, q_3)$$

$$R_6^3(\vartheta_4, \vartheta_5, \vartheta_6) = R_3^{0T} R = \begin{bmatrix} n_x^3 & s_x^3 & a_x^3 \\ n_y^3 & s_y^3 & a_y^3 \\ n_z^3 & s_z^3 & a_z^3 \end{bmatrix}$$

$$T_6^3(q) = A_4^3 A_5^4 A_6^5 = \begin{bmatrix} c_4 c_5 c_6 - s_4 s_6 & -c_4 c_5 s_6 - s_4 c_6 & c_4 s_5 & c_4 s_5 d_6 \\ s_4 c_5 c_6 + c_4 s_6 & -s_4 c_5 s_6 + c_4 c_6 & s_4 s_5 & s_4 s_5 d_6 \\ -s_5 c_6 & s_5 s_6 & c_5 & c_5 d_6 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

YZZ Euler set

Admits two solutions as seen before



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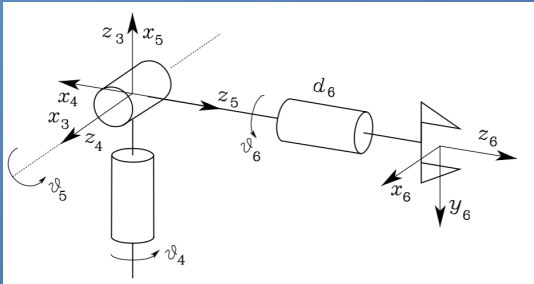
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Inverse Kinematics

Solution of the Anthropomorphic arm with spherical wrist:



6 – Solve for $\vartheta_4, \vartheta_5, \vartheta_6$

$$T_6^3(q) = A_4^3 A_5^4 A_6^5 = \begin{bmatrix} c_4 c_5 c_6 - s_4 s_6 & -c_4 c_5 s_6 - s_4 c_6 & c_4 s_5 & c_4 s_5 d_6 \\ s_4 c_5 c_6 + c_4 s_6 & -s_4 c_5 s_6 + c_4 c_6 & s_4 s_5 & s_4 s_5 d_6 \\ -s_5 c_6 & s_5 s_6 & c_5 & c_5 d_6 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

$$R_6^3(\vartheta_4, \vartheta_5, \vartheta_6) = R_3^{0T} R = \begin{bmatrix} n_x^3 & s_x^3 & a_x^3 \\ n_y^3 & s_y^3 & a_y^3 \\ n_z^3 & s_z^3 & a_z^3 \end{bmatrix}$$

$$\vartheta_5 \in]0, \pi[$$

$$\begin{aligned} \vartheta_4 &= \text{Atan2}(a_y^3, a_x^3) \\ \vartheta_5 &= \text{Atan2}\left(\sqrt{(a_x^3)^2 + (a_y^3)^2}, a_z^3\right) \\ \vartheta_6 &= \text{Atan2}(s_z^3, -n_z^3) \end{aligned}$$

$$\vartheta_5 \in]-\pi, 0[$$

$$\begin{aligned} \vartheta_4 &= \text{Atan2}(-a_y^3, -a_x^3) \\ \vartheta_5 &= \text{Atan2}\left(-\sqrt{(a_x^3)^2 + (a_y^3)^2}, a_z^3\right) \\ \vartheta_6 &= \text{Atan2}(-s_z^3, n_z^3) \end{aligned}$$



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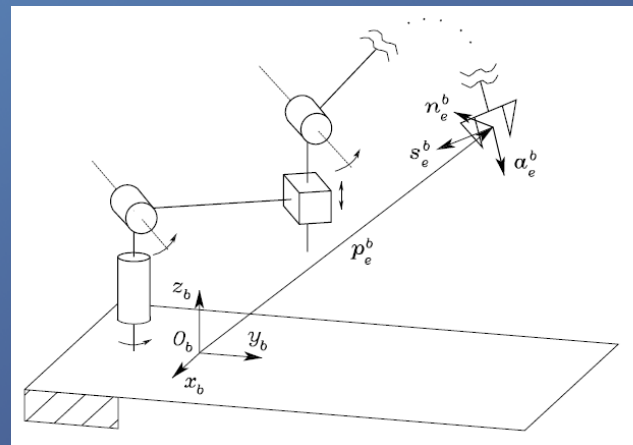
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Differential Kinematics

The geometric Jacobian

Kinematics

$$T_e(q) = \begin{bmatrix} R_e(q) & p_e(q) \\ 0^T & 1 \end{bmatrix}$$



Differential Kinematics

$$v_e = \begin{bmatrix} \dot{p}_e \\ \omega_e \end{bmatrix} = J(q) \dot{q}$$

Geometric Jacobian

$$J = \begin{bmatrix} J_P \\ J_O \end{bmatrix}$$

Angular Velocity

$$\dot{R} = S(\omega) R$$



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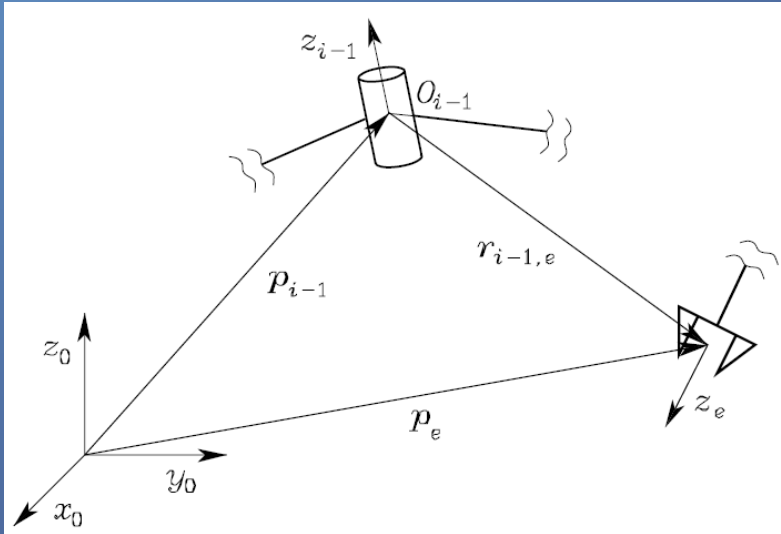
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Differential Kinematics

Calculation of the geometric Jacobian: **Linear velocity**



$$\dot{p}_e = \sum_{i=1}^n \frac{\partial p_e}{\partial q_i} \dot{q}_i = \sum_{i=1}^n J_{Pi} \dot{q}_i$$

Prismatic Joint

$$\dot{q}_i J_{Pi} = \dot{d}_i z_{i-1}$$

$$J_{Pi} = z_{i-1}$$

Revolute Joint

$$\dot{q}_i J_{Pi} = \dot{\vartheta}_i z_{i-1} \times (p_e - p_{i-1})$$

$$J_{Pi} = z_{i-1} \times (p_e - p_{i-1})$$



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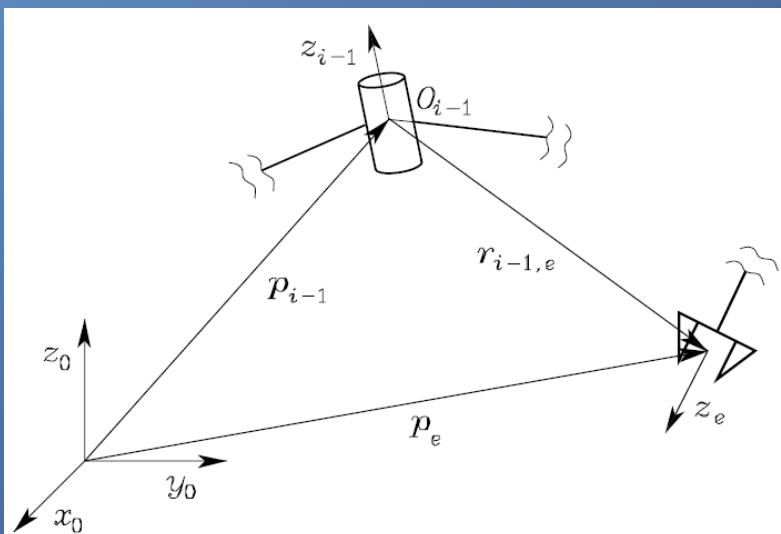
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Differential Kinematics

Calculation of the geometric Jacobian: **Angular velocity**



$$\omega_e = \omega_n = \sum_{i=1}^n \omega_{i-1,i} = \sum_{i=1}^n J_{Oi} \dot{q}_i$$

Prismatic Joint

$$\dot{q}_i J_{Oi} = 0$$

$$J_{Oi} = 0$$

Revolute Joint

$$\dot{q}_i J_{Oi} = \dot{\vartheta}_i z_{i-1}$$

$$J_{Oi} = z_{i-1}$$



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Differential Kinematics

Calculation of the geometric Jacobian

$$J = \begin{bmatrix} J_{P1} & \dots & J_{Pn} \\ J_{O1} & \dots & J_{On} \end{bmatrix}$$

$$\begin{bmatrix} J_{Pi} \\ J_{Oi} \end{bmatrix} = \begin{cases} \begin{bmatrix} z_{i-1} \\ 0 \end{bmatrix} \\ \begin{bmatrix} z_{i-1} \times (p_e - p_{i-1}) \\ z_{i-1} \end{bmatrix} \end{cases}$$

Prismatic Joint

Revolute Joint

$$z_{i-1} = R_1^0(q_1) \dots R_{i-1}^{i-2}(q_{i-1}) z_0$$

$$\tilde{p}_e = A_1^0(q_1) \dots A_n^{n-1}(q_n) \tilde{p}_0$$

$$\tilde{p}_{i-1} = A_1^0(q_1) \dots A_{i-1}^{i-2}(q_{i-1}) \tilde{p}_0$$

$$z_0 = \begin{bmatrix} 0 & 0 & 1 \end{bmatrix}^T$$

$$\tilde{p}_0 = \begin{bmatrix} 0 & 0 & 0 & 1 \end{bmatrix}^T$$



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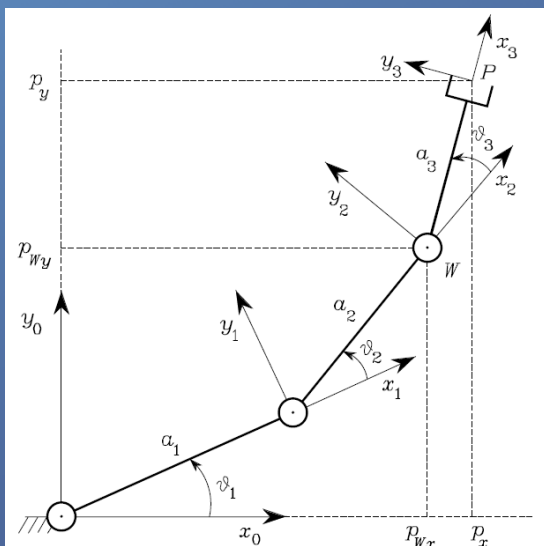
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Differential Kinematics

Example: Three-link planar arm

$$J(q) = \begin{bmatrix} z_0 \times (p_3 - p_0) & z_1 \times (p_3 - p_1) & z_2 \times (p_3 - p_2) \\ z_0 & z_1 & z_2 \end{bmatrix}$$



From the D-H kinematics

$$p_0 = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix} \quad p_1 = \begin{bmatrix} a_1 c_1 \\ a_1 s_1 \\ 0 \end{bmatrix} \quad p_2 = \begin{bmatrix} a_1 c_1 + a_2 c_{12} \\ a_1 s_1 + a_2 s_{12} \\ 0 \end{bmatrix}$$

$$p_3 = \begin{bmatrix} a_1 c_1 + a_2 c_{12} + a_3 c_{123} \\ a_1 s_1 + a_2 s_{12} + a_3 s_{123} \\ 0 \end{bmatrix}$$

$$z_0 = z_1 = z_2 = \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}$$



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Kinematic Singularities

The Jacobian is, in general, a function of the configuration $\mathbf{q}$. Those configurations at which $\mathbf{J}$ is rank-deficient are termed kinematic singularities.

- Singularities represent **configurations at which mobility of the structure is reduced**, i.e., it is not possible to impose an arbitrary motion to the end-effector.
- When the structure is at a singularity, **infinite solutions to the inverse kinematics problem may exist**.
- In the neighborhood of a singularity, **small velocities in the operational space may cause large velocities in the joint space**.



Kinematic Singularities

- **Boundary singularities** that occur when the manipulator is either outstretched or retracted.
 - They can be avoided on condition that the manipulator is not driven to the boundaries of its reachable workspace.
- **Internal singularities** that occur inside the reachable workspace and are generally caused by the alignment of two or more axes of motion, or else by the attainment of particular end-effector configurations.
 - These constitute a serious problem, as they can be encountered anywhere in the reachable workspace for a planned path in the operational space.



Kinematic Singularities

Singularity Decoupling:

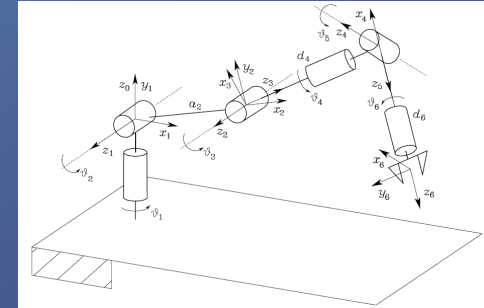
For manipulators having a spherical wrist it is possible to split the problem of singularity computation into two separate problems

- computation of **arm singularities**
- computation of **wrist singularities**

$$J = \begin{bmatrix} J_{11} & J_{12} \\ J_{21} & J_{22} \end{bmatrix}$$

with $p = p_w$

$$J_{12} = \begin{bmatrix} 0 & 0 & 0 \end{bmatrix}$$



$$\det(J) = \det(J_{11})\det(J_{22})$$



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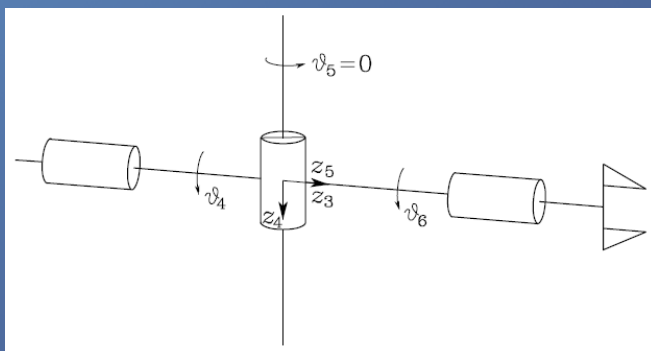
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Kinematic Singularities

Wrist Singularitys: spherical wrist

$$J_{22} = \begin{bmatrix} z_3 & z_4 & z_5 \end{bmatrix}$$



If $\theta_5=0$ or $\theta_5=\pi$, z_3 and z_5 are linearly dependant

This singularity can be encountered anywhere in the reachable workspace!



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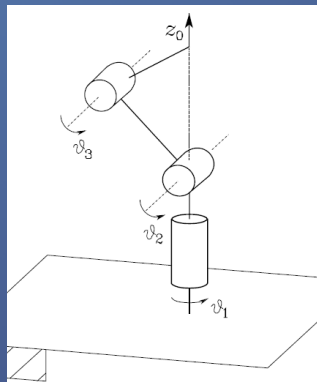
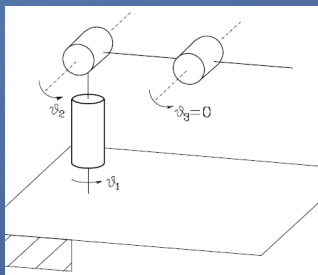
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Kinematic Singularities

Arm singularities: anthropomorphic arm

$$\det(\mathbf{J}_{11}) = -a_2 a_3 s_3 (a_2 c_2 + a_3 c_{23})$$



$$\det(\mathbf{J}_{11}) = 0$$

$$\text{if } \theta_3 = 0 \text{ or } \theta_3 = \pi \\ \text{or } a_2 c_2 + a_3 c_{23} = 0$$

Clearly identified in the robot workspace:
can be avoided!



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Kinematic Redundancy

n : DOFs of the structure

m : operational space variables

r : operational space variables necessary for the task ($r \leq m$)

$$\mathbf{v}_e = \begin{bmatrix} \dot{\mathbf{p}}_e \\ \boldsymbol{\omega}_e \end{bmatrix} = \mathbf{J}(\mathbf{q}) \dot{\mathbf{q}}$$

$(r \times 1)$ $(r \times n)$ $(n \times 1)$

If $r < n$ the manipulator is kinematically redundant

Functionally redundant if $r < m$

Intrinsically redundant $r = m$



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Statics

Relationship between the generalized forces applied to the end-effector and the generalized forces applied to the joints at equilibrium

τ : the $(n \times 1)$ vector of joint torques

γ : the $(r \times 1)$ vector of end-effector forces

$$dW_\tau = \tau^T dq$$

$$\begin{aligned} dW_\gamma &= f_e^T J_P(q) dq + \mu_e^T J_O(q) dq \\ &= \gamma_e^T J(q) dq \end{aligned}$$



$$\tau = J^T(q) \gamma_e$$

The relationship between the end-effector forces and the joint torques is established by the transpose of the manipulator geometric Jacobian!



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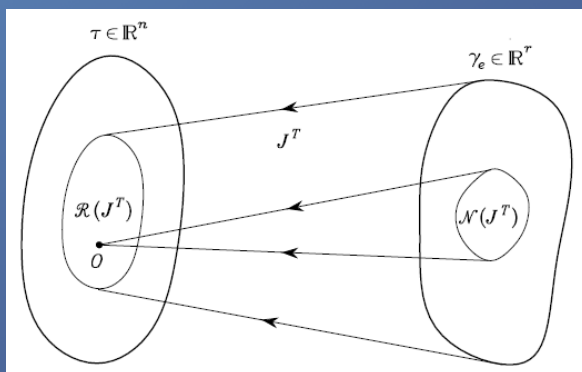
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Statics

- The range space of J^T is the subspace $\mathcal{R}(J^T)$ in $\mathbb{R}^n$ of the joint torques that can balance the end-effector forces, in the given manipulator posture.
- The null space of J^T is the subspace $\mathcal{N}(J^T)$ in $\mathbb{R}^r$ of the end-effector forces that do not require any balancing joint torques, in the given manipulator posture.



End-effector forces

$$\gamma_e \in \mathcal{N}(J^T)$$

are entirely
absorbed by the structure



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Inverse Kinematics Estimation

Inverse Kinematics problem consists of **computing joint trajectories $q(t)$ to be used as reference** inputs to the joint servos

Admits closed-form solutions only for manipulators having a simple kinematic structure.

These limitations are caused by the **highly nonlinear relationship between joint space variables and operational space variables.**

However, Differential Kinematics represents a **linear mapping between the joint velocity space and the operational velocity space**, although it is dependent of the joint configuration.

$$v_e = \begin{bmatrix} \dot{p}_e \\ \omega_e \end{bmatrix} = J(q)\dot{q}$$



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Inverse Kinematics Estimation

If J is full rank ($n = r$) joint velocities can be obtained from

$$\dot{q} = J^{-1}(q)v_e$$

In **discrete time**, the computation of joint velocity yields the relation

$$q(t_{k+1}) = q(t_k) + \dot{q}(t_k)\Delta t$$

That allows the estimation of the **next** joint position from the current joint position and the prescribed end-effector velocity.

$$q(t_{k+1}) = q(t_k) + J^{-1}(q(t_k))v_e(t_k)\Delta t$$

Dynamics!

Quality of estimation depends on:



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Inverse Kinematics Estimation

Closed Loop Inverse Kinematics (CLIK)

$$\dot{q} = J^{-1}(q) \begin{bmatrix} \dot{p}_d + K_P e_P \\ \omega_d + K_O e_O \end{bmatrix}$$

Proportional
Feedback control of
the estimation!

$$e_P = p_d - p_e(q)$$

$$e_O = r \vartheta$$

with

$$R(\vartheta, r) = R_d R_e^T(q)$$

K_P and K_O are user selected proportional feedback gain matrices



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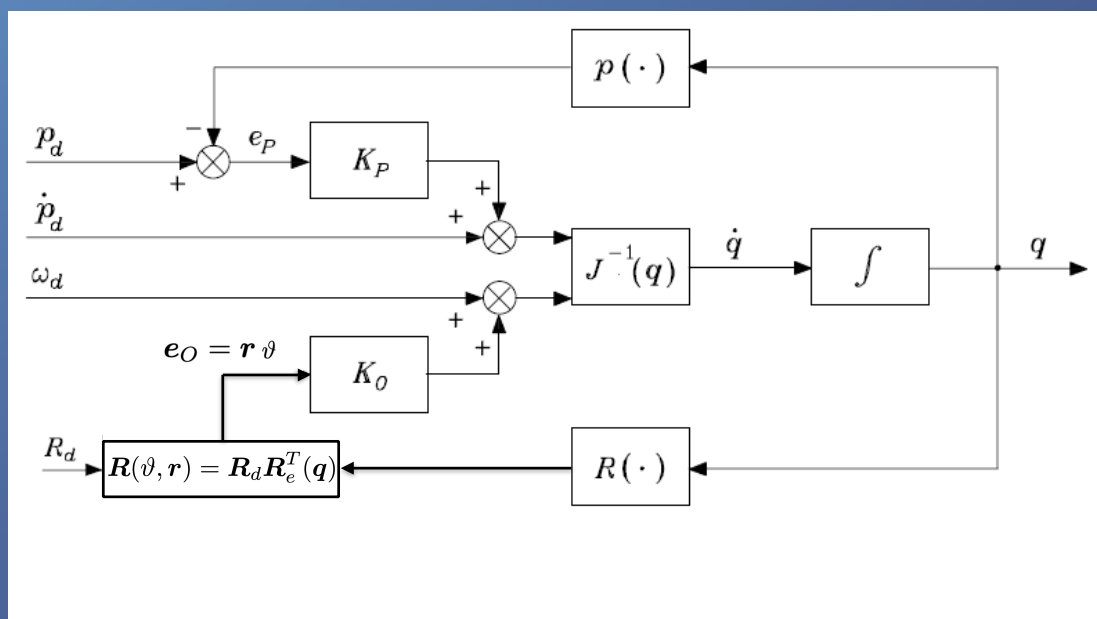
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Inverse Kinematics Estimation

Closed Loop Inverse Kinematics (CLIK)



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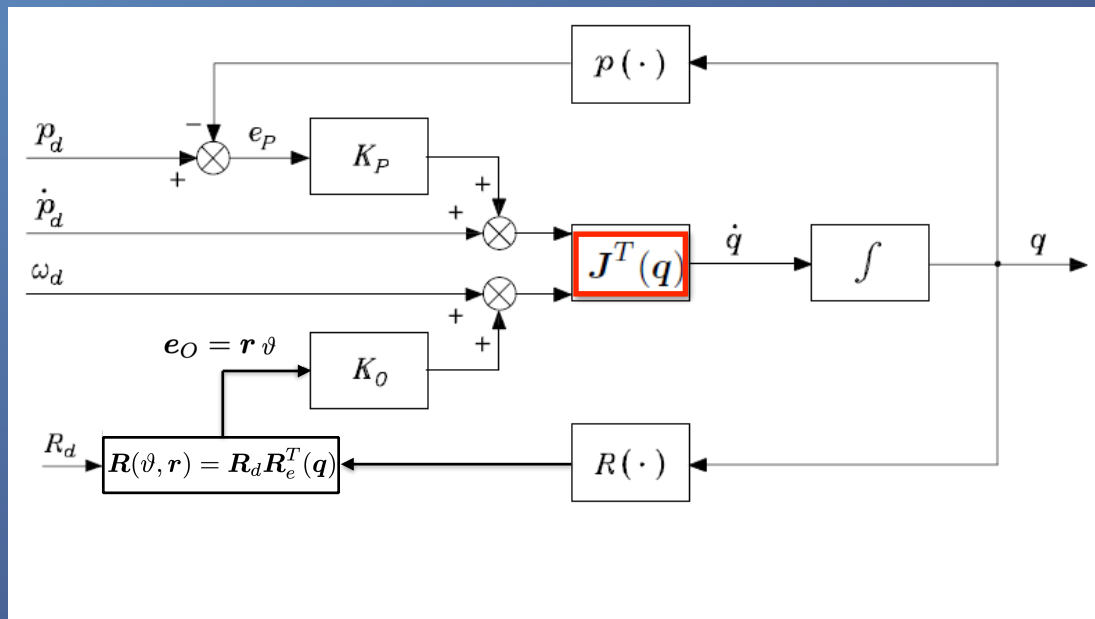
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Inverse Kinematics Estimation

Closed Loop Inverse Kinematics (CLIK) - transposed based



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Inverse Kinematics Estimation

Closed Loop Inverse Kinematics (CLIK) Redundant Manipulators

$$\dot{q} = J^\dagger v_e$$

Right pseudo-inverse

$$J^\dagger = J^T (J J^T)^{-1}$$

Stabilization of the null-space

$$\dot{q} = J^\dagger v_e + (I_n - J^\dagger J) \dot{q}_0$$



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Inverse Kinematics Estimation

Closed Loop Inverse Kinematics (CLIK)

Redundant Manipulators

Stabilization of the null-space

$$\dot{\mathbf{q}} = \mathbf{J}^\dagger \mathbf{v}_e + (\mathbf{I}_n - \mathbf{J}^\dagger \mathbf{J}) \dot{\mathbf{q}}_0$$

$$\dot{\mathbf{q}}_0 = k_0 \left(\frac{\partial w(\mathbf{q})}{\partial \mathbf{q}} \right)^T$$

Maximize distance from mechanical joint limits:

$$w(\mathbf{q}) = -\frac{1}{2n} \sum_{i=1}^n \left(\frac{q_i - \bar{q}_i}{q_{iM} - q_{im}} \right)^2$$



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Inverse Kinematics Estimation

Closed Loop Inverse Kinematics (CLIK)

Redundant Manipulators

Kinematic Singularities

$$\dot{\mathbf{q}} = \mathbf{J}^\dagger \mathbf{v}_e + (\mathbf{I}_n - \mathbf{J}^\dagger \mathbf{J}) \dot{\mathbf{q}}_0$$

$$\mathbf{J}^\dagger = \mathbf{J}^T (\mathbf{J} \mathbf{J}^T)^{-1}$$

Damped Least Squares Inverse

$$\mathbf{J}^\star = \mathbf{J}^T (\mathbf{J} \mathbf{J}^T + k^2 \mathbf{I})^{-1}$$



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Trajectory Planning

Goal: To generate the reference inputs to the motion control system which ensures that the manipulator executes the planned trajectories.

Trajectory planning in the joint space

Trajectory planning in the operational space

Point-to-point motion

Motion through a sequence of points

Path

The locus of points in the joint space, or in the operational space, which the manipulator has to follow in the execution of the assigned motion

≠

Trajectory

A path on which a timing law is specified.



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Trajectory planning in the joint space

Point-to-point motion – Polynomial trajectory profiles

Knowing the initial $(q_i, \dot{q}_i)$ and final $(q_f, \dot{q}_f, t_f)$ joint positions and velocities determine the coefficients of the *cubic polynomial*:

$$q(t) = a_3 t^3 + a_2 t^2 + a_1 t + a_0$$

For the velocities and accelerations results

$$\dot{q}(t) = 3a_3 t^2 + 2a_2 t + a_1$$

$$\ddot{q}(t) = 6a_3 t + 2a_2$$

and the trajectory coefficients are obtained by solving the boundary equations:

$$\begin{aligned} a_0 &= q_i \\ a_1 &= \dot{q}_i \\ a_3 t_f^3 + a_2 t_f^2 + a_1 t_f + a_0 &= q_f \\ 3a_3 t_f^2 + 2a_2 t_f + a_1 &= \dot{q}_f \end{aligned}$$



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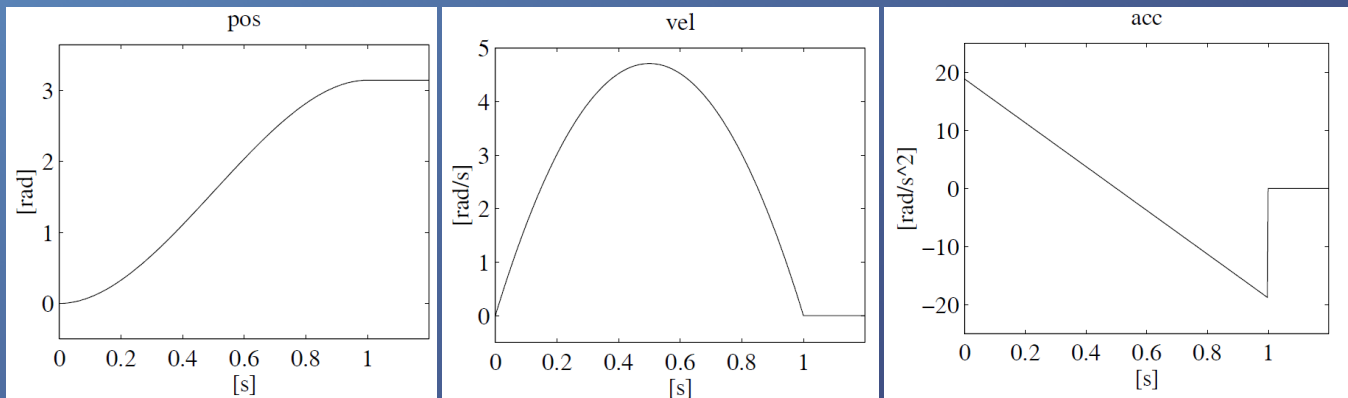
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Trajectory planning in the joint space

Point-to-point motion – Polynomial trajectory profiles

$$q(t) = a_3 t^3 + a_2 t^2 + a_1 t + a_0$$

$$q_i = 0, q_f = \pi, t_f = 1, \text{ and } \dot{q}_i = \dot{q}_f = 0$$



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Trajectory planning in the joint space

Point-to-point motion – Polynomial trajectory profile

Knowing the initial $(q_i, \dot{q}_i, \ddot{q}_i)$ and final $(q_f, \dot{q}_f, \ddot{q}_f, t_f)$ joint positions, velocities and accelerations determine the coefficients of the *fifth order polynomial*:

$$q(t) = a_5 t^5 + a_4 t^4 + a_3 t^3 + a_2 t^2 + a_1 t + a_0$$

The same steps as for the cubic polynomial hold.

Higher order polynomials may be considered



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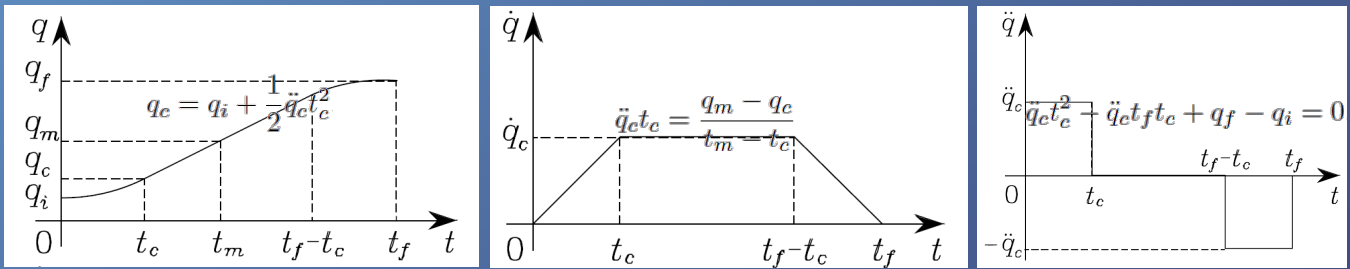
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Trajectory planning in the joint space

Point-to-point motion – Blended Polynomial trajectory profiles

When one wishes to easily verify the velocities and accelerations of the trajectory against the maximum allowed by the manipulator, blended polynomials are preferred:



Knowing the initial (q_i) and final (q_f, t_f) joint positions and the transition velocity ($\dot{q}_c$) determine the three blended *polynomials*:

$$q(t) = \begin{cases} q_i + \frac{1}{2}\ddot{q}_c t^2 & 0 \leq t \leq t_c \\ q_i + \dot{q}_c t_c(t - t_c/2) & t_c < t \leq t_f - t_c \\ q_f - \frac{1}{2}\ddot{q}_c (t_f - t)^2 & t_f - t_c < t \leq t_f. \end{cases}$$

with

$$t_c = \frac{q_i - q_f + \dot{q}_c t_f}{\dot{q}_c}$$

$$\ddot{q}_c = \frac{\dot{q}_c^2}{q_i - q_f + \dot{q}_c t_f}$$



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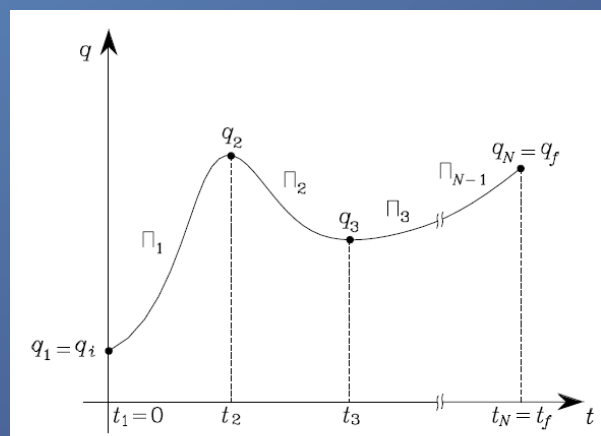
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Trajectory planning in the joint space

Motion through a sequence of points

In many practical applications, it is necessary to describe the path with a number of points greater than two:



The problem is now to generate a trajectory where N points, termed *path points*, have to be reached by the manipulator at certain instants of time.



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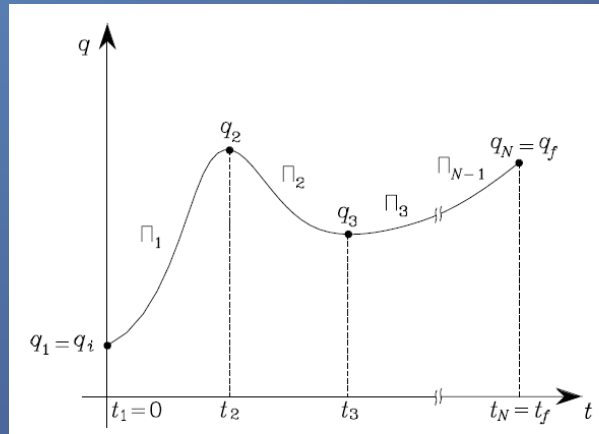
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Trajectory planning in the joint space

Motion through a sequence of points

The solution is to choose $N-1$ cubic polynomials continuous at the path points



The following situations can be considered:

1. Interpolating polynomials with imposed velocities at path points;
2. Interpolating polynomials with continuous accelerations at path points (splines)



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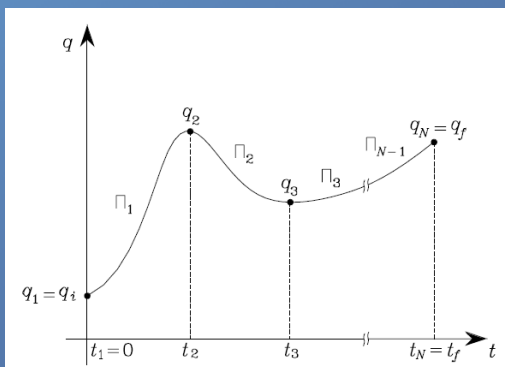
Robótica de Manipulação

Prof. Jorge M. M. Martins

Trajectory planning in the joint space

Motion through a sequence of points

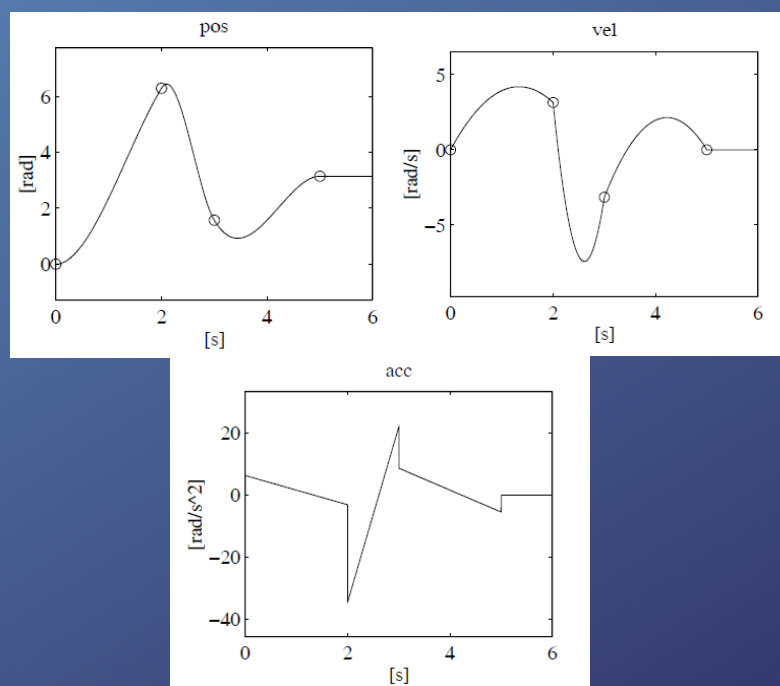
1. Interpolating polynomials with imposed velocities at path points;



for $k = 1, \dots, N-1$:

$$\begin{aligned} \Pi_k(t_k) &= q_k \\ \Pi_k(t_{k+1}) &= q_{k+1} \\ \dot{\Pi}_k(t_k) &= \dot{q}_k \\ \dot{\Pi}_k(t_{k+1}) &= \dot{q}_{k+1} \end{aligned}$$

$$\dot{\Pi}_k(t_{k+1}) = \dot{\Pi}_{k+1}(t_{k+1})$$



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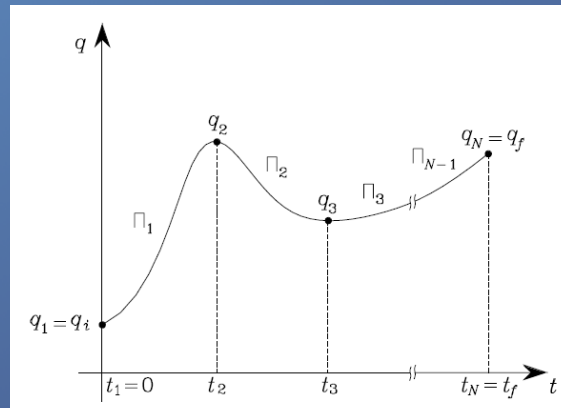
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Trajectory planning in the joint space

Motion through a sequence of points

2. Interpolating polynomials with continuous accelerations at path points (splines)



$$\begin{aligned} \Pi_{k-1}(t_k) &= q_k \\ \dot{\Pi}_{k-1}(t_k) &= \dot{\Pi}_k(t_k) \\ \ddot{\Pi}_{k-1}(t_k) &= \ddot{\Pi}_k(t_k) \\ \ddot{\Pi}_{k-1}(t_k) &= \ddot{\Pi}_k(t_k). \end{aligned}$$



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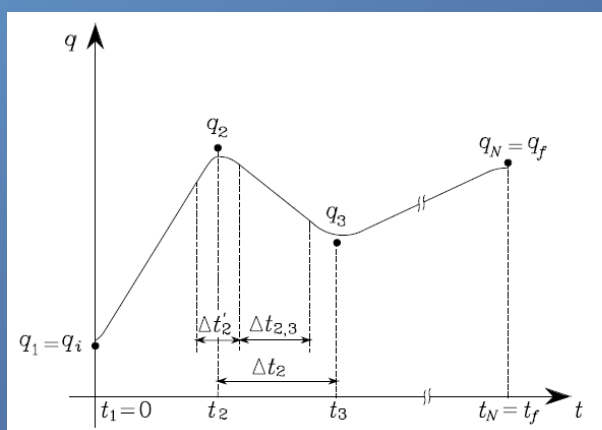
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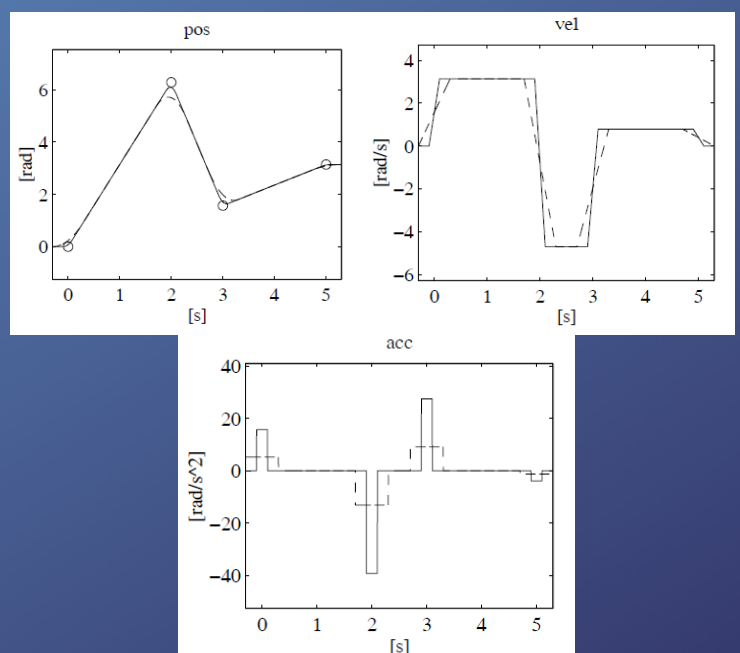
Trajectory planning in the joint space

Motion through a sequence of points

Approximating linear polynomials with parabolic blends



$$\begin{aligned} \dot{q}_{k-1,k} &= \frac{q_k - q_{k-1}}{\Delta t_{k-1}} \\ \ddot{q}_k &= \frac{\dot{q}_{k,k+1} - \dot{q}_{k-1,k}}{\Delta t'_k}; \end{aligned}$$



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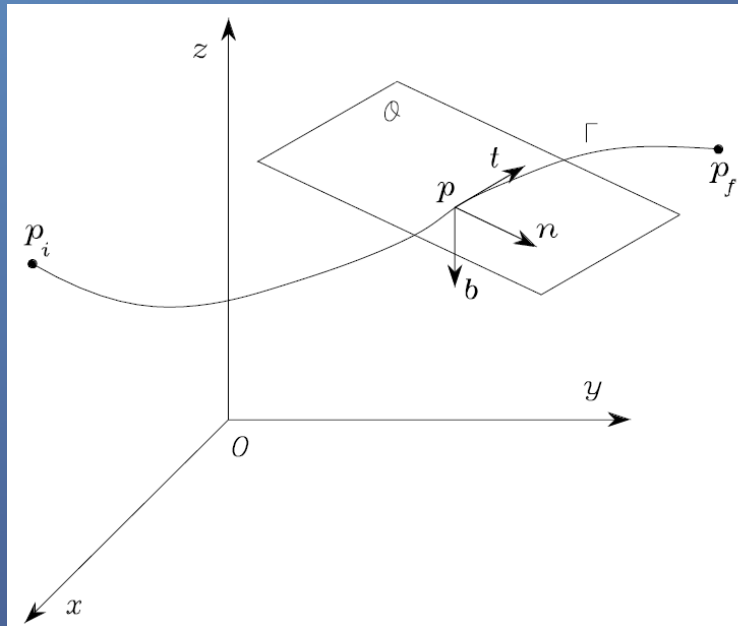
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Trajectory planning in the operational space

Analytical trajectories – parametric curves

Parametric description of trajectories in space

- Motion Primitives – geometric features of the path
- Time Primitives – timing law on the path



$$p = f(s)$$

s – Arc Length

Path reference frame

$$t = \frac{dp}{ds}$$

$$n = \frac{1}{\left\| \frac{d^2 p}{ds^2} \right\|} \frac{d^2 p}{ds^2}$$

$$b = t \times n.$$



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Robótica de Manipulação

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Trajectory planning in the operational space

Analytical trajectories – Rectilinear path

$$p(s) = p_i + \frac{s}{\|p_f - p_i\|} (p_f - p_i)$$

$$p(0) = p_i \text{ and } p(\|p_f - p_i\|) = p_f$$

$$\frac{dp}{ds} = \frac{1}{\|p_f - p_i\|} (p_f - p_i)$$

$$\frac{d^2 p}{ds^2} = 0.$$

For a rectilinear path it is not possible to define the path reference frame uniquely.

Timing law for manipulator end effector position

$$\dot{p}_e = \dot{s} \frac{dp_e}{ds} = \dot{s} t$$

$$\dot{p}_e = \frac{\dot{s}}{\|p_f - p_i\|} (p_f - p_i) = \dot{s} t$$

$$\ddot{p}_e = \frac{\ddot{s}}{\|p_f - p_i\|} (p_f - p_i) = \ddot{s} t.$$



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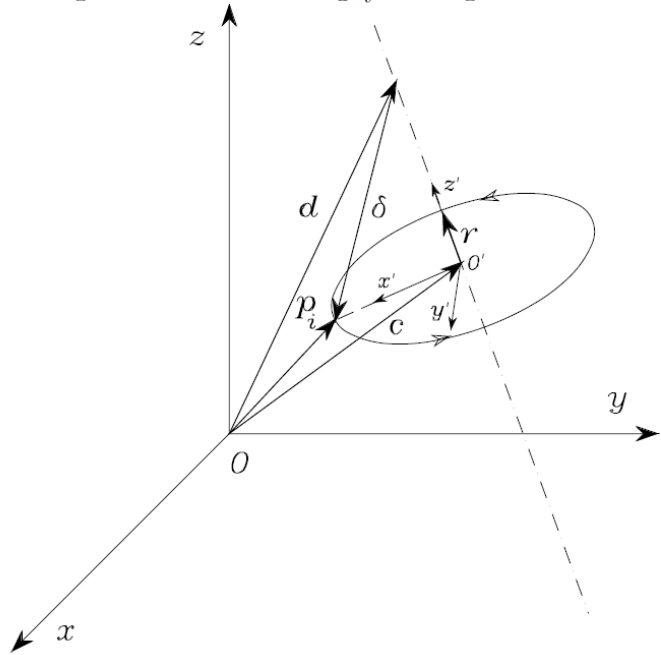
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Trajectory planning in the operational space

Analytical trajectories – Circular path

the unit vector of the circle axis $\mathbf{r}$,
the position vector $\mathbf{d}$ of a point along the circle axis,
the position vector $\mathbf{p}_i$ of a point on the circle.



c – center of the circle

$$\mathbf{c} = \mathbf{d} + (\delta^T \mathbf{r}) \mathbf{r}$$

ρ – radius of the circle

$$\rho = \|\mathbf{p}_i - \mathbf{c}\|$$

Circle reference frame

$$\mathbf{R} = [\mathbf{x}' \quad \mathbf{y}' \quad \mathbf{z}']$$

Parametric representation

$$\mathbf{p}'(s) = \begin{bmatrix} \rho \cos(s/\rho) \\ \rho \sin(s/\rho) \\ 0 \end{bmatrix}$$

$$\mathbf{p}(s) = \mathbf{c} + \mathbf{R} \mathbf{p}'(s)$$



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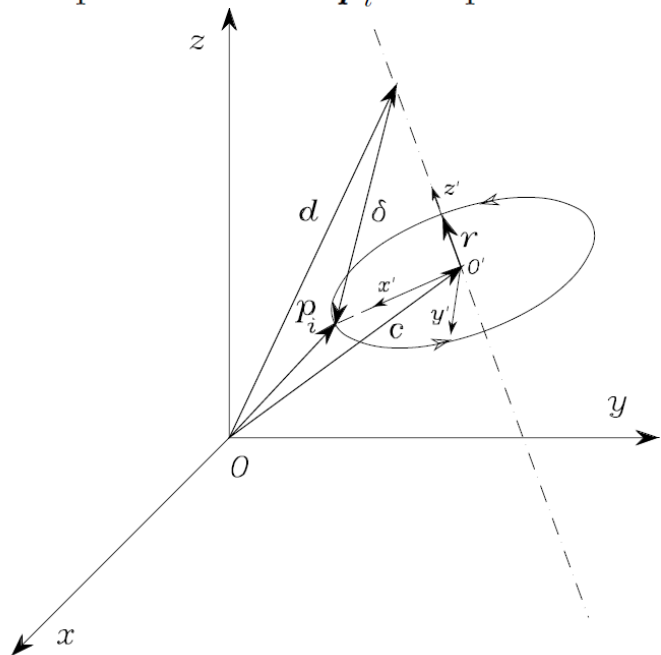
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the position vector $\mathbf{p}_i$ of a point on the circle.



Parametric representation

$$\mathbf{p}(s) = \mathbf{c} + \mathbf{R} \mathbf{p}'(s)$$

$$\frac{d\mathbf{p}}{ds} = \mathbf{R} \begin{bmatrix} -\sin(s/\rho) \\ \cos(s/\rho) \\ 0 \end{bmatrix}$$

$$\frac{d^2\mathbf{p}}{ds^2} = \mathbf{R} \begin{bmatrix} -\cos(s/\rho)/\rho \\ -\sin(s/\rho)/\rho \\ 0 \end{bmatrix}$$



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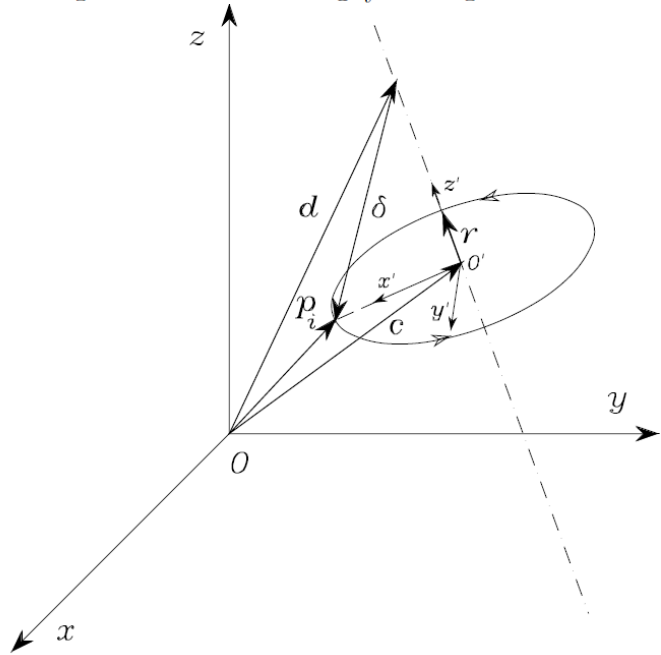
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Timing Law

$$\dot{\mathbf{p}}_e = \dot{s} \frac{d\mathbf{p}_e}{ds} = \dot{s} \mathbf{t}$$

$$\dot{\mathbf{p}}_e = \mathbf{R} \begin{bmatrix} -\dot{s} \sin(s/\rho) \\ \dot{s} \cos(s/\rho) \\ 0 \end{bmatrix}$$

$$\ddot{\mathbf{p}}_e = \mathbf{R} \begin{bmatrix} -\dot{s}^2 \cos(s/\rho)/\rho - \ddot{s} \sin(s/\rho) \\ -\dot{s}^2 \sin(s/\rho)/\rho + \ddot{s} \cos(s/\rho) \\ 0 \end{bmatrix}$$



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Trajectory planning in the operational space

Analytical trajectories – Orientation

Euler Angles

$$\phi_e = (\varphi, \vartheta, \psi)$$

Linear segment

$$\phi_e = \phi_i + \frac{s}{\|\phi_f - \phi_i\|} (\phi_f - \phi_i)$$

$$\dot{\phi}_e = \frac{\dot{s}}{\|\phi_f - \phi_i\|} (\phi_f - \phi_i)$$

$$\ddot{\phi}_e = \frac{\ddot{s}}{\|\phi_f - \phi_i\|} (\phi_f - \phi_i);$$



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Trajectory planning in the operational space

Analytical trajectories – Orientation

Angle and Axis

Given R_i and R_f at t_f :

$$R_f^i = R_i^T R_f = \begin{bmatrix} r_{11} & r_{12} & r_{13} \\ r_{21} & r_{22} & r_{23} \\ r_{31} & r_{32} & r_{33} \end{bmatrix}$$

$$\vartheta_f = \cos^{-1} \left(\frac{r_{11} + r_{22} + r_{33} - 1}{2} \right)$$
$$\mathbf{r} = \frac{1}{2 \sin \vartheta_f} \begin{bmatrix} r_{32} - r_{23} \\ r_{13} - r_{31} \\ r_{21} - r_{12} \end{bmatrix}$$

The matrix $R^i(t)$ with $R^i(0)=I$ and $R^i(t_f)=R_f^i$ can be interpreted as the matrix $R^i(\vartheta(t), \mathbf{r}^i)$, a *single joint problem*, where it suffices to assign a timing law to $\vartheta(t)$: $\vartheta(0)=0$ and $\vartheta(t_f)=\vartheta_f$ with constant $\mathbf{r}^i$

$$\begin{aligned} \boldsymbol{\omega}^i &= \dot{\vartheta} \mathbf{r}^i \\ \dot{\boldsymbol{\omega}}^i &= \ddot{\vartheta} \mathbf{r}^i. \end{aligned}$$

$$\begin{aligned} R_e(t) &= R_i R^i(t) \\ \boldsymbol{\omega}_e(t) &= R_i \boldsymbol{\omega}^i(t) \\ \dot{\boldsymbol{\omega}}_e(t) &= R_i \dot{\boldsymbol{\omega}}^i(t). \end{aligned}$$



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